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(54) **SAWHORSE**

(71) Applicants: **KETER HOME AND GARDEN PRODUCTS LTD**, Herzliya (IL); **MILWAUKEE ELECTRIC TOOL CORPORATION**, Brookfield, WI (US)

(72) Inventors: **Yaron Brunner**, Gvat (IL); **Izhar Shani, I**, Gvat (IL); **Uri Parizer**, Hukok (IL)

(73) Assignee: **KETER HOME AND GARDEN PRODUCTS LTD**, Herzliya (IL)

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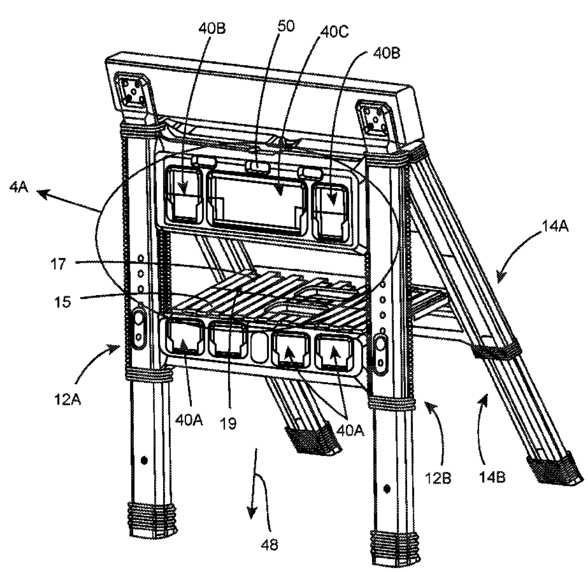
Primary Examiner — Brian D Keller
Assistant Examiner — Jason Khalil Hawkins
(74) *Attorney, Agent, or Firm* — Cantor Colburn LLP

(57) **ABSTRACT**

The present disclosure concerns a sawhorse. More particularly the disclosure concerns a sawhorse configured with an articulation system.

21 Claims, 19 Drawing Sheets



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E1—“Furniture Mechanics”, 1993, pp. 124-125 and Machine translation to English.
E2—“Advanced Manufacturing Technology”, 2005, p. 59 and Machine translation to English.

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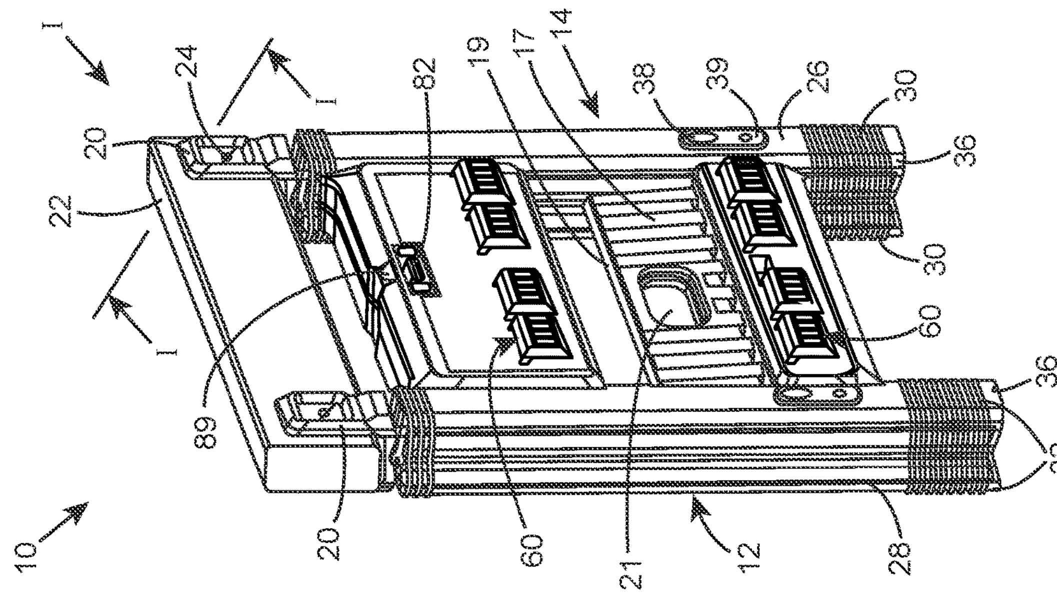
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FIG. 1B

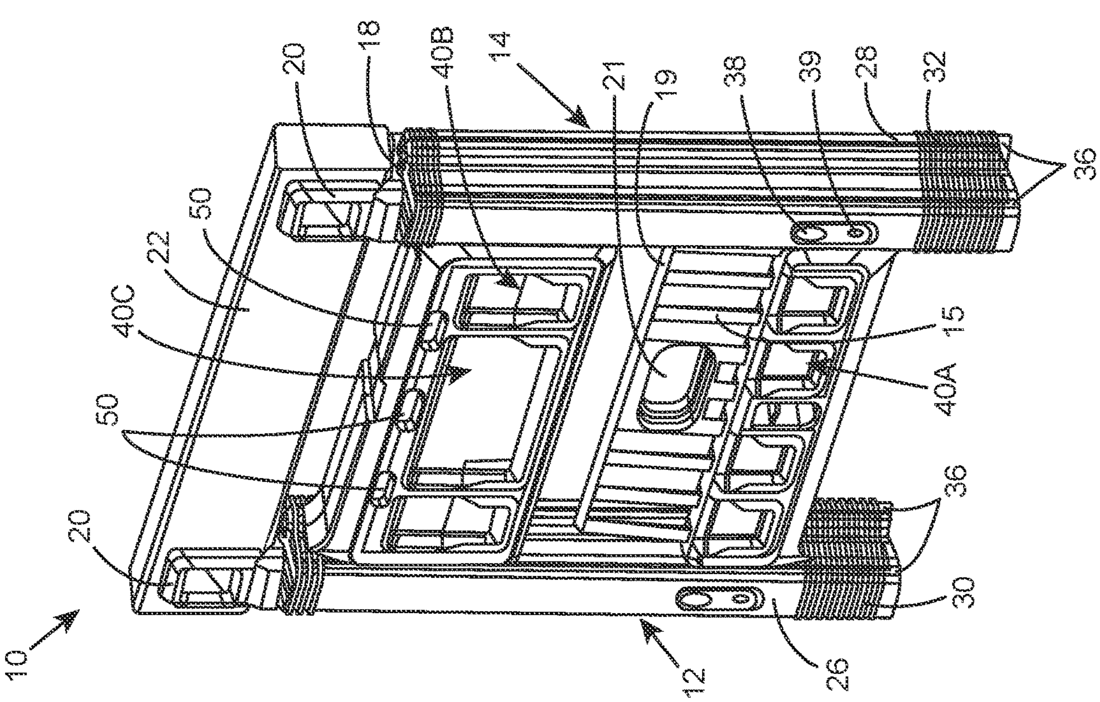
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FIG. 1A

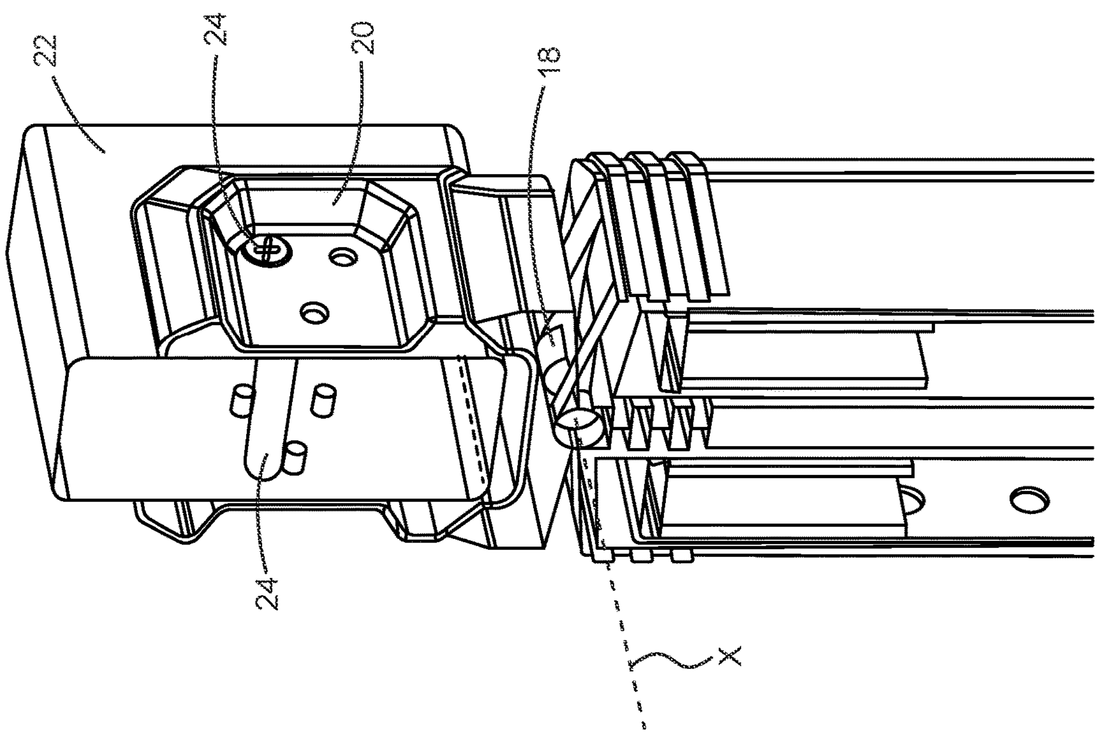

FIG. 1D

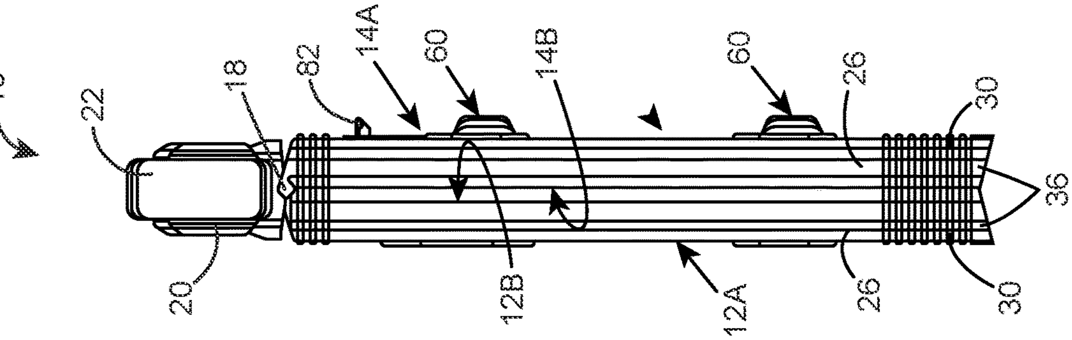

FIG. 1C

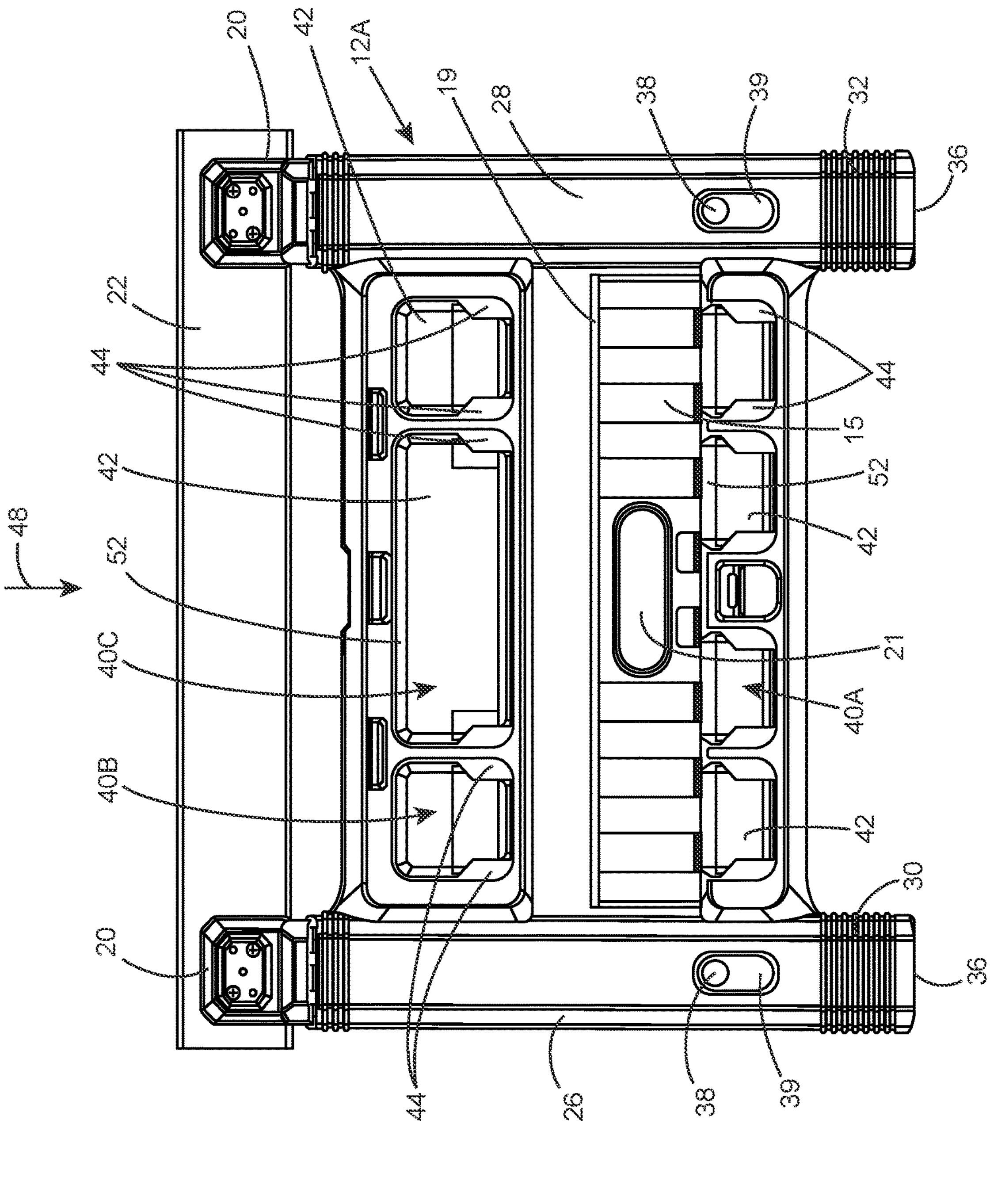
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FIG. 2A

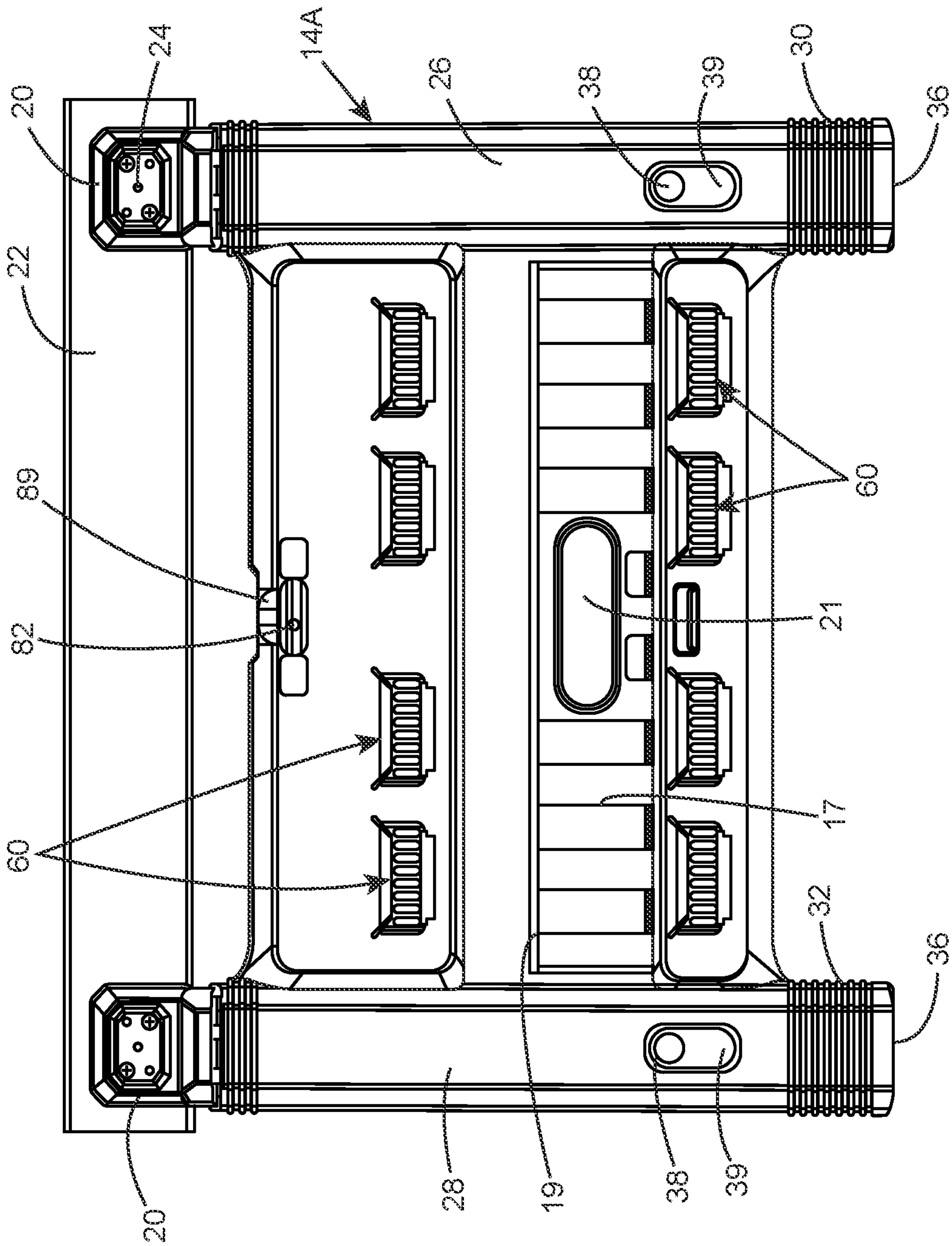

FIG. 2B

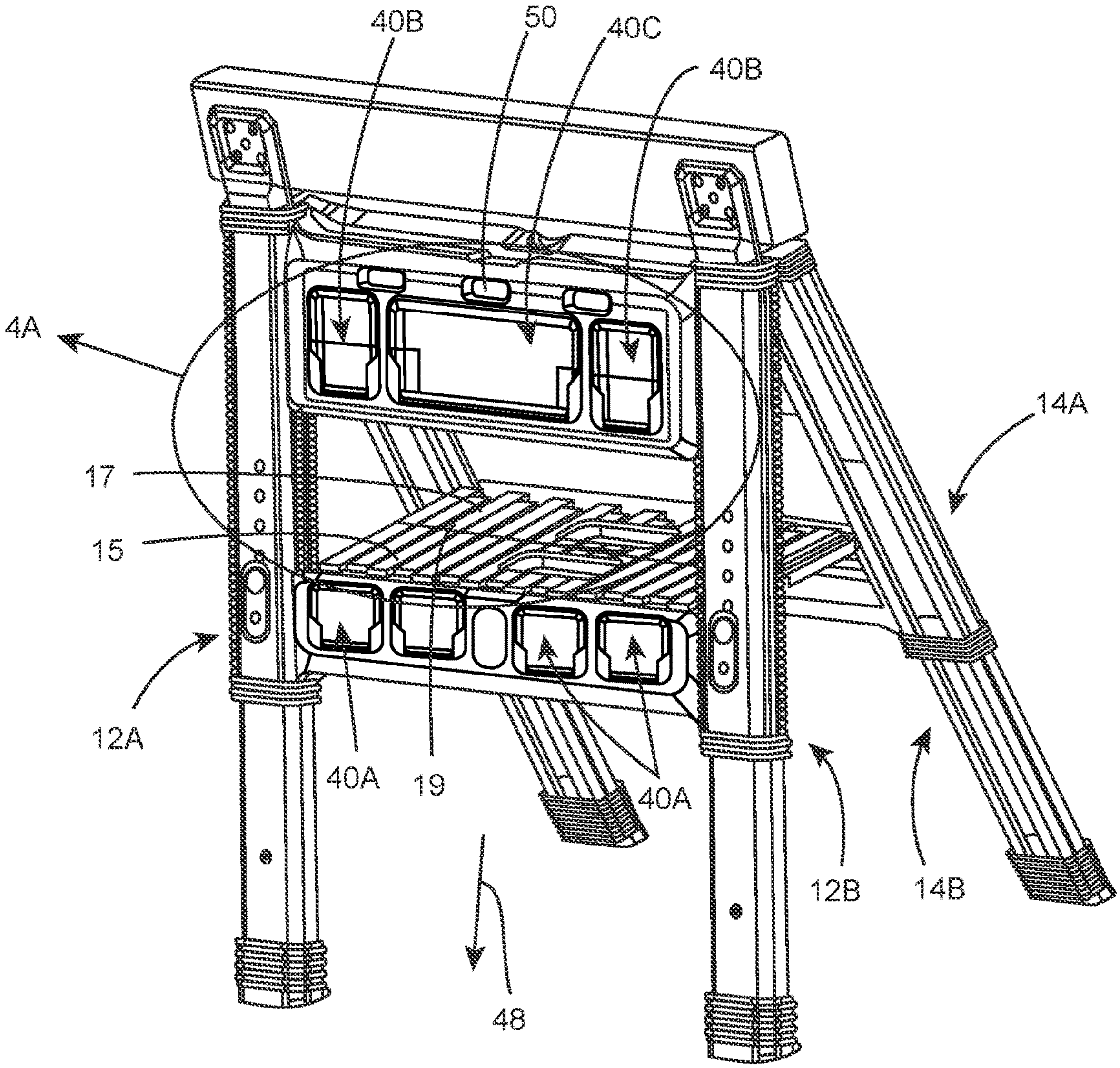

FIG. 3A

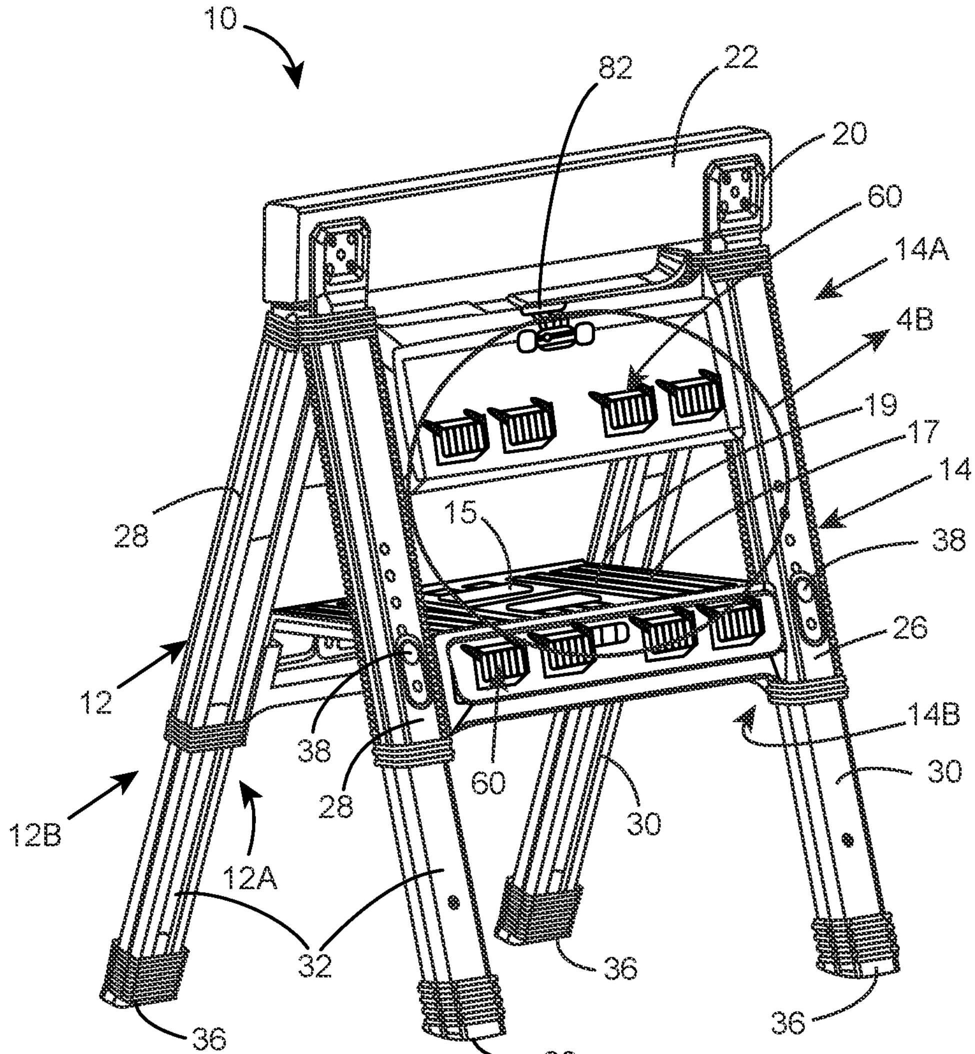
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FIG. 3B

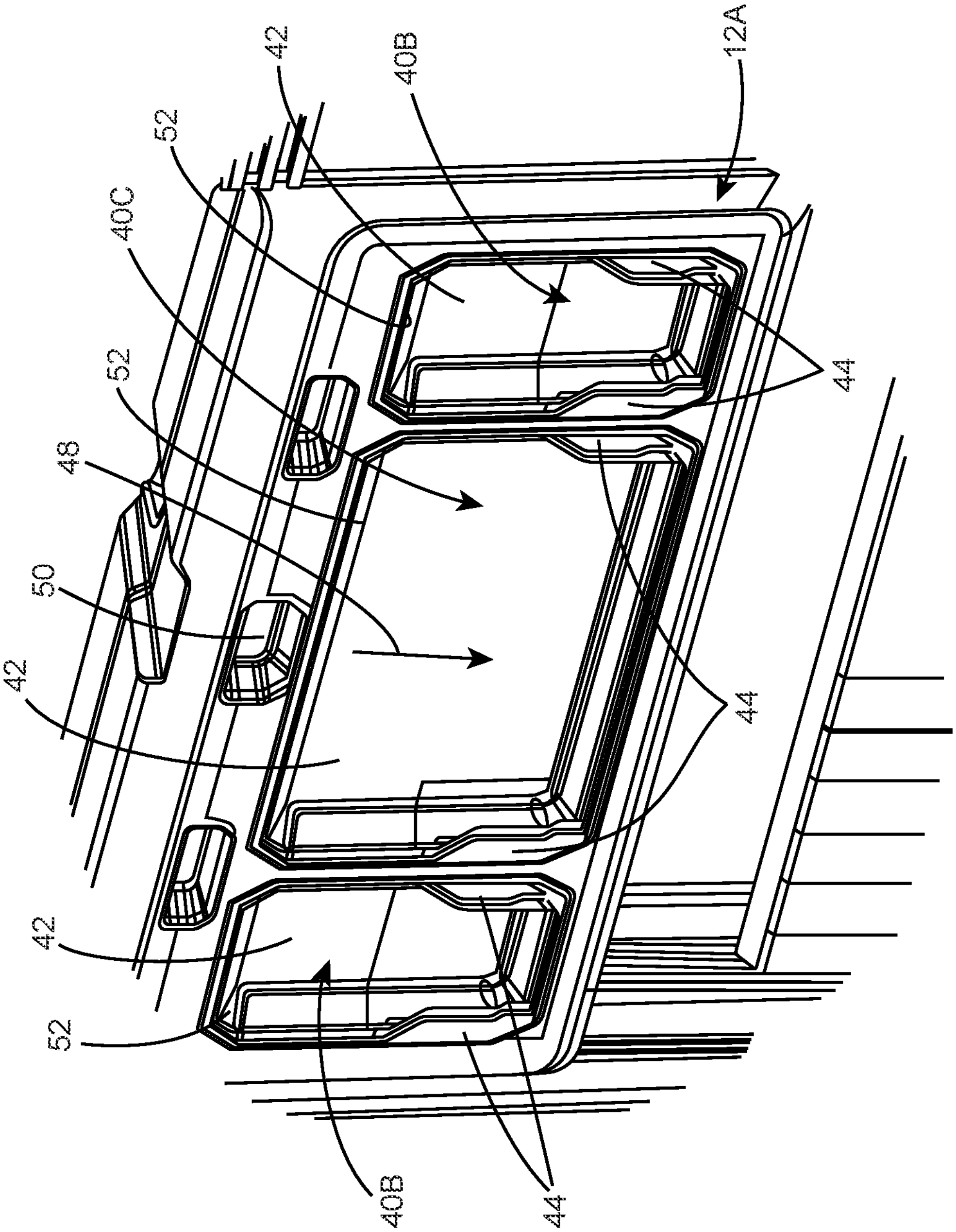

FIG. 4A

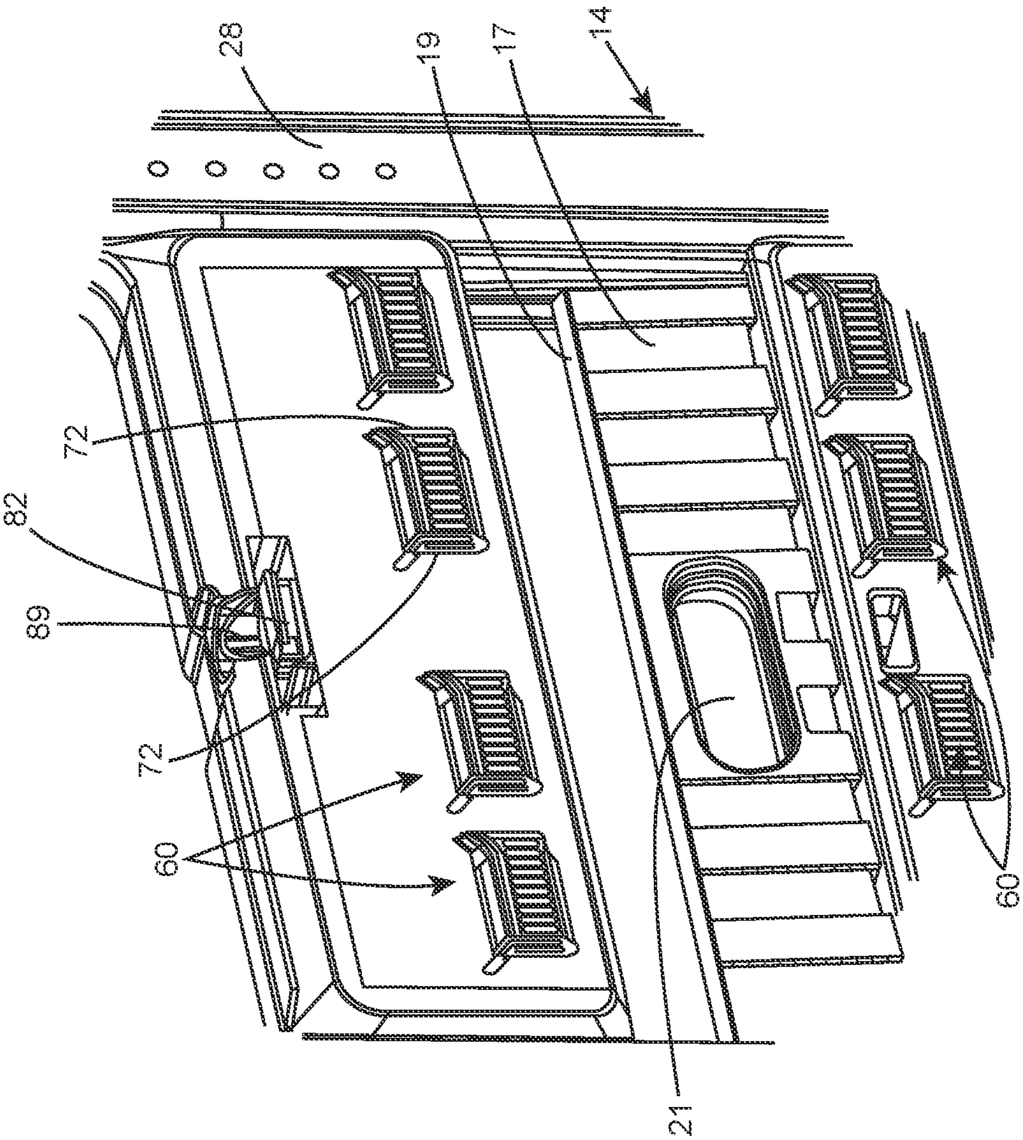

FIG. 4B

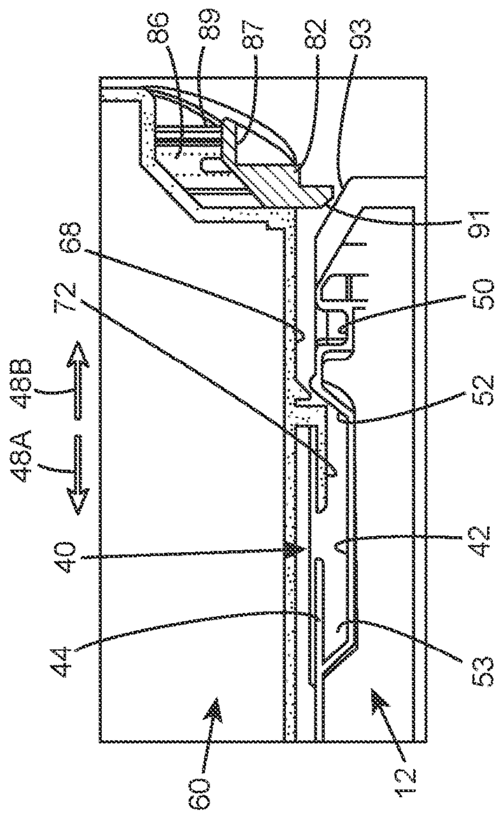

FIG. 5A

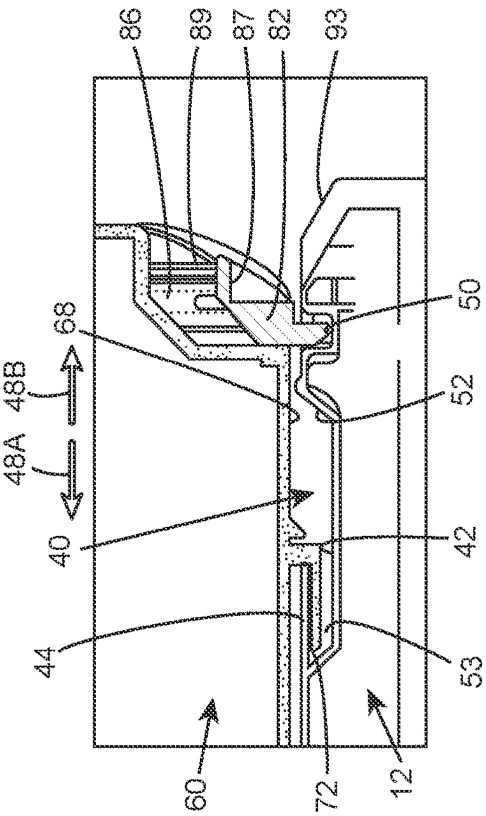

FIG. 5B

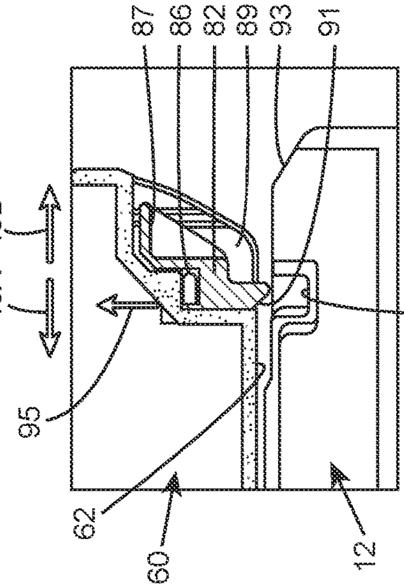

FIG. 6A

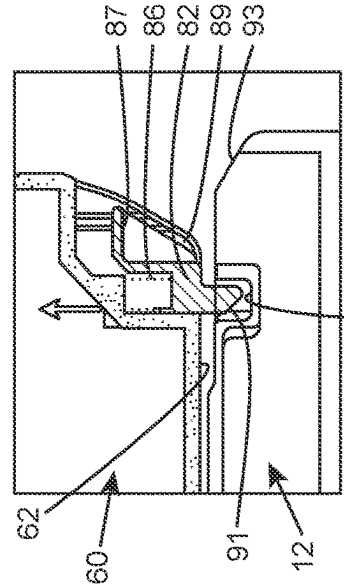

FIG. 6B

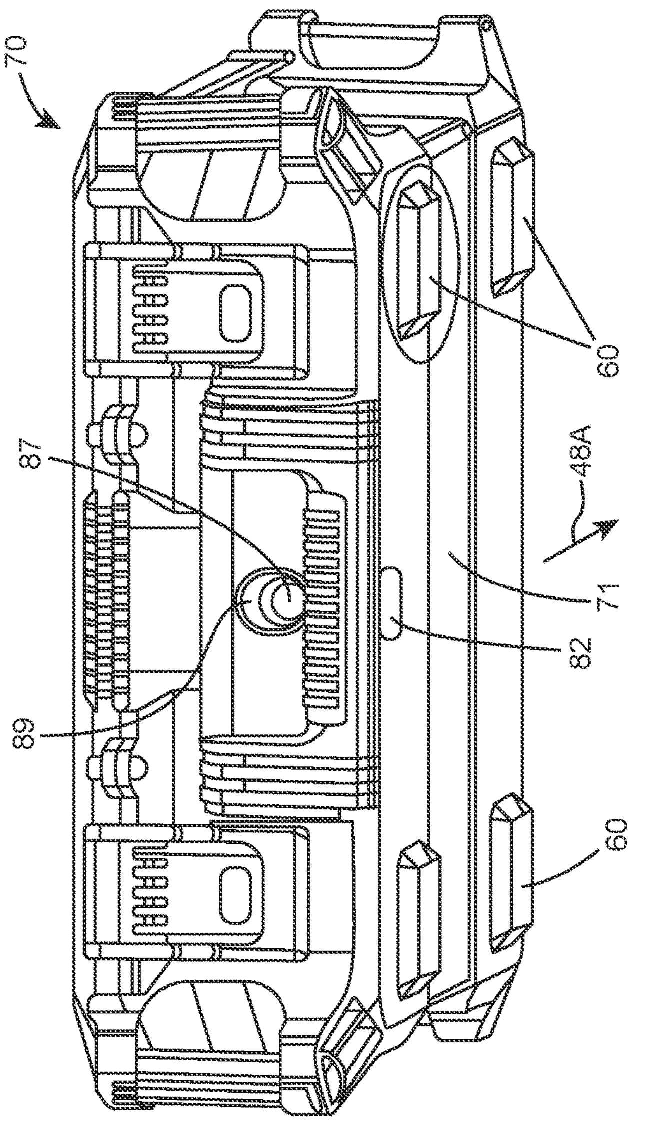

FIG. 7

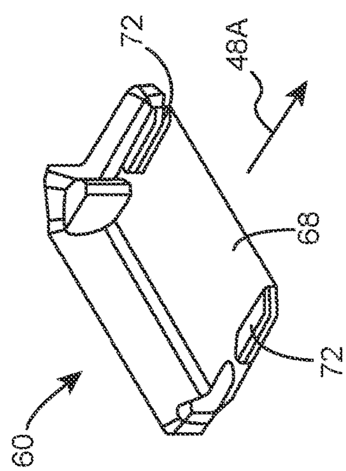

FIG. 8A

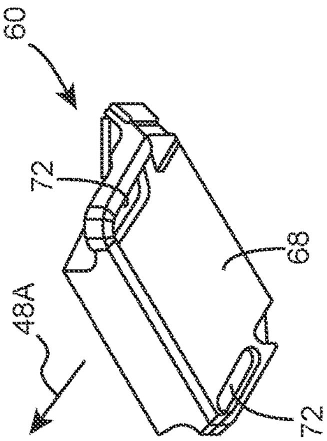

FIG. 8B

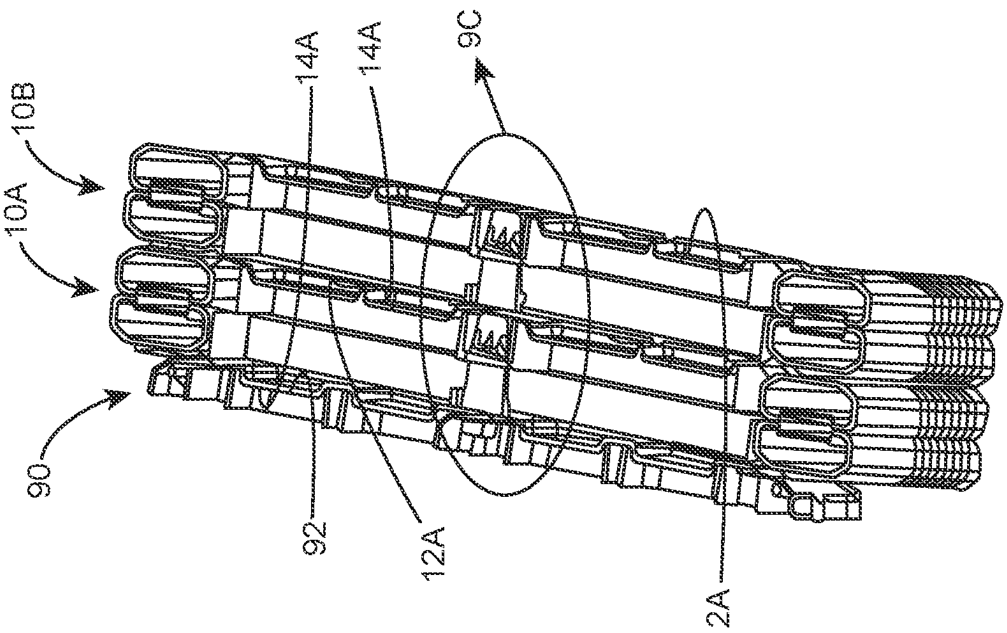

FIG. 9B

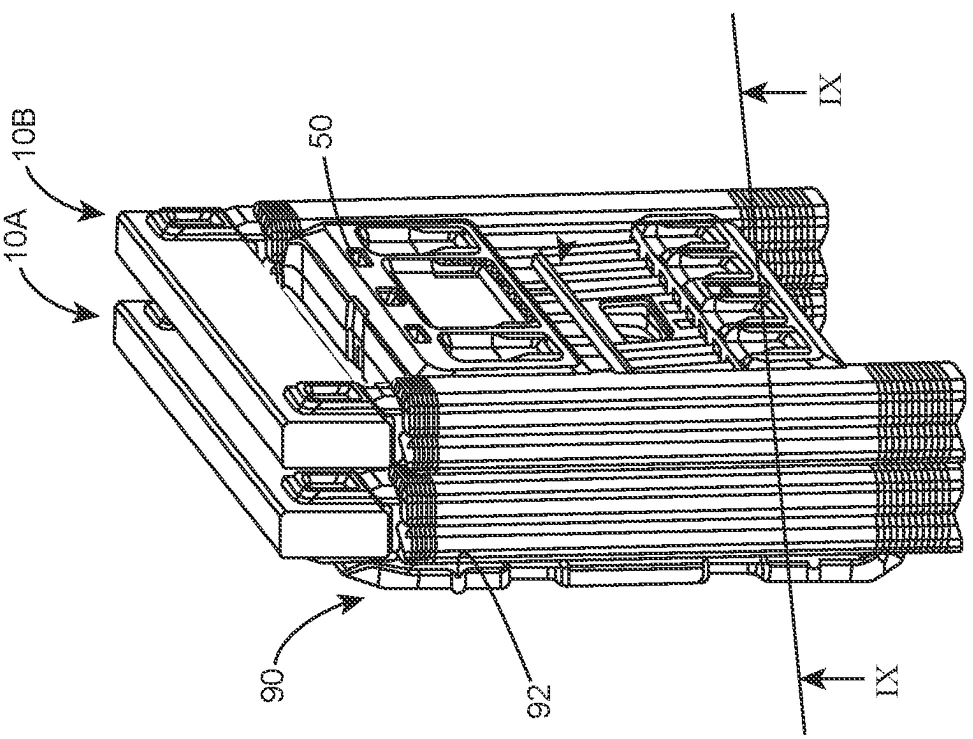

FIG. 9A

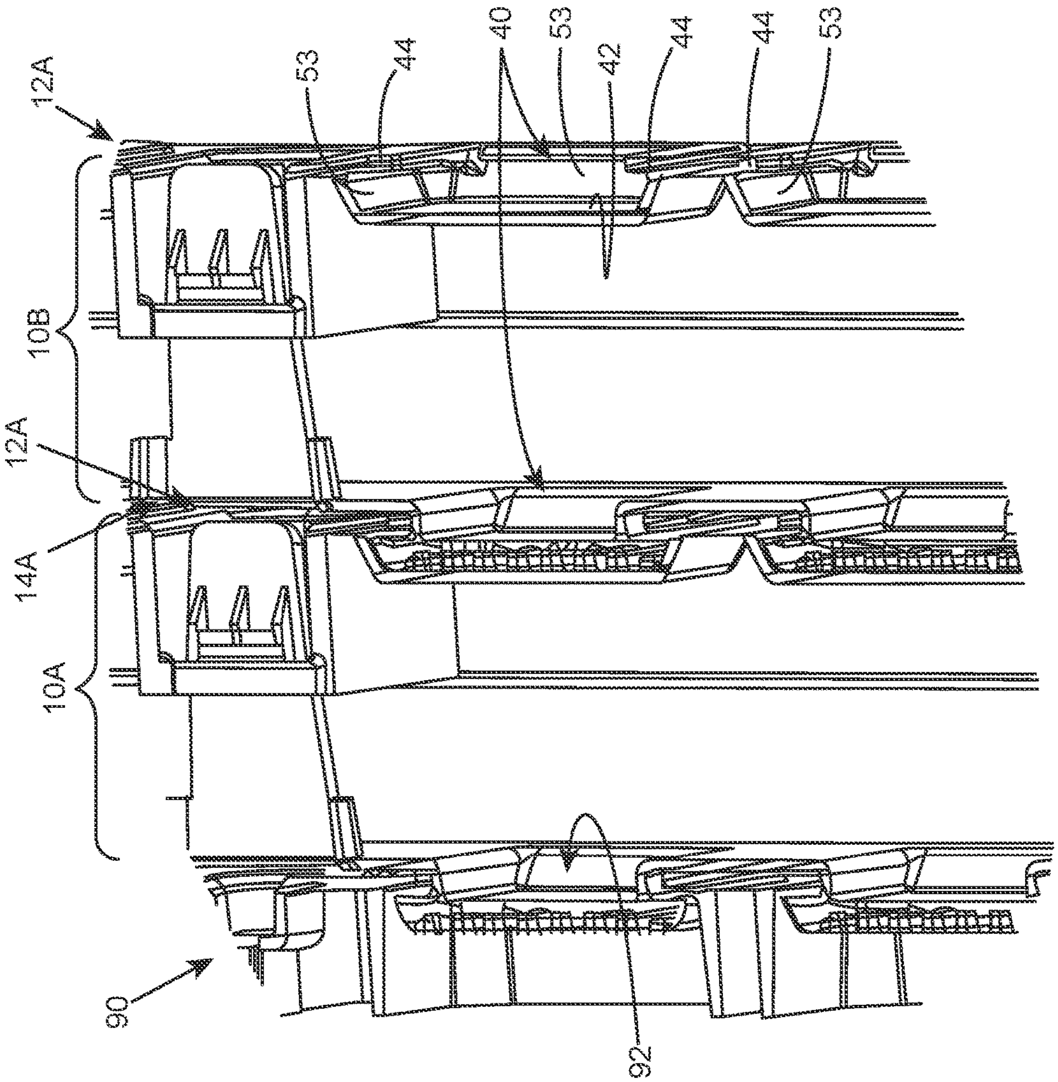

FIG. 9C

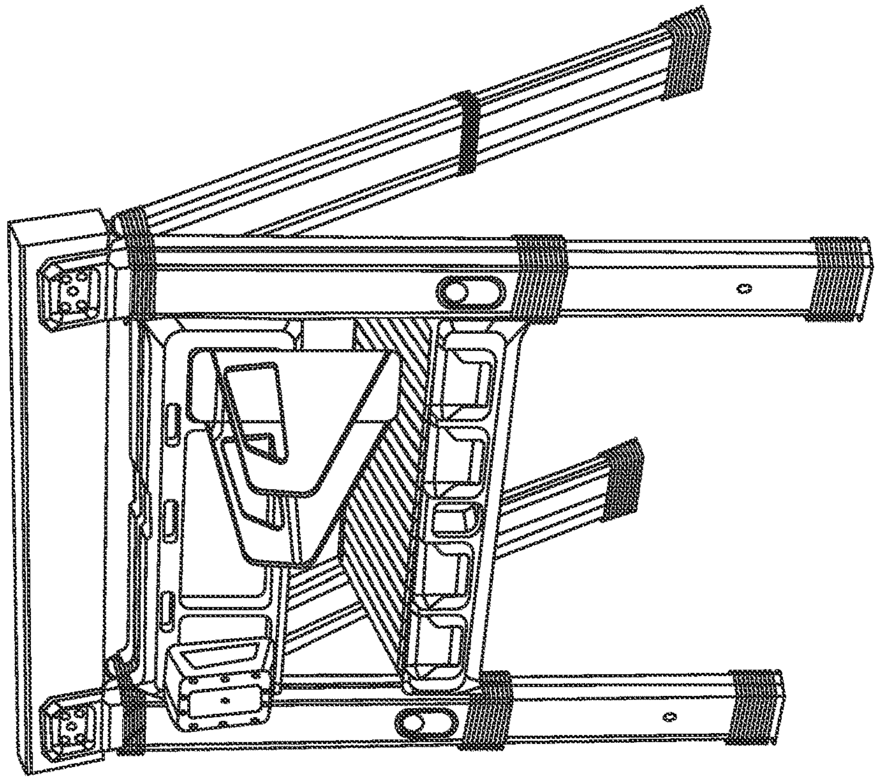

FIG. 10B

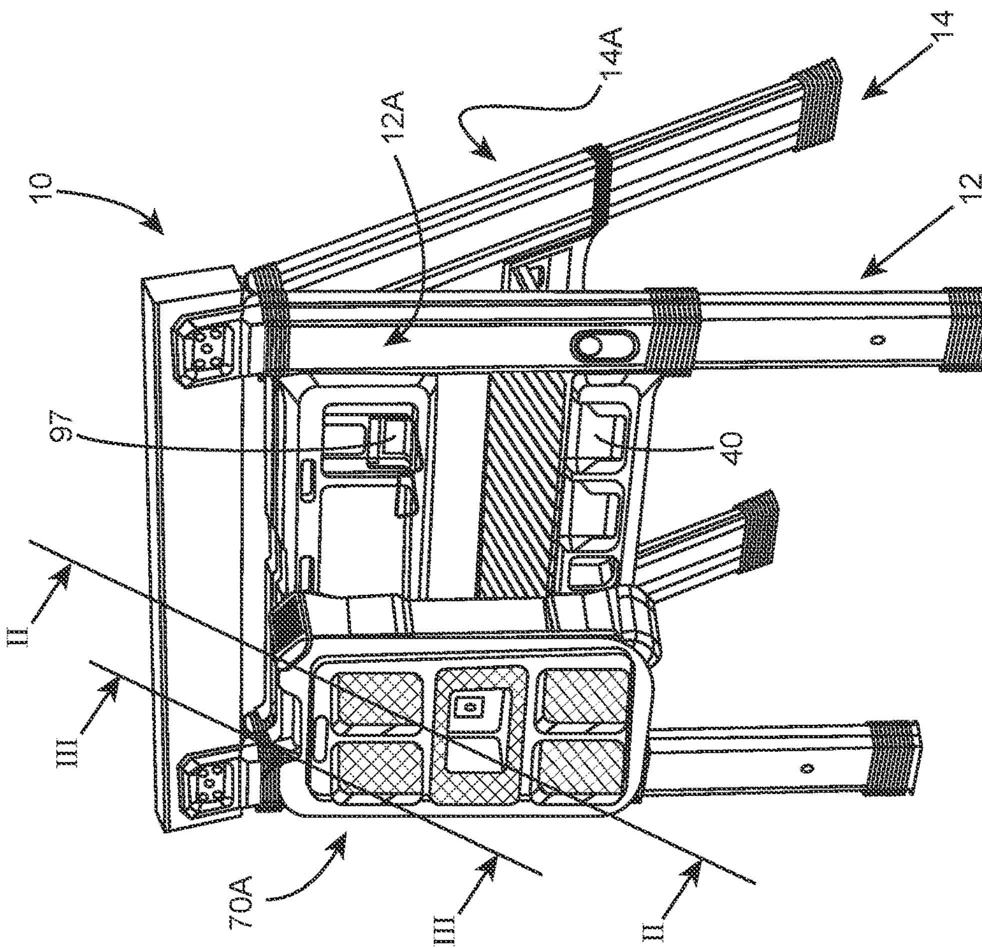

FIG. 10A

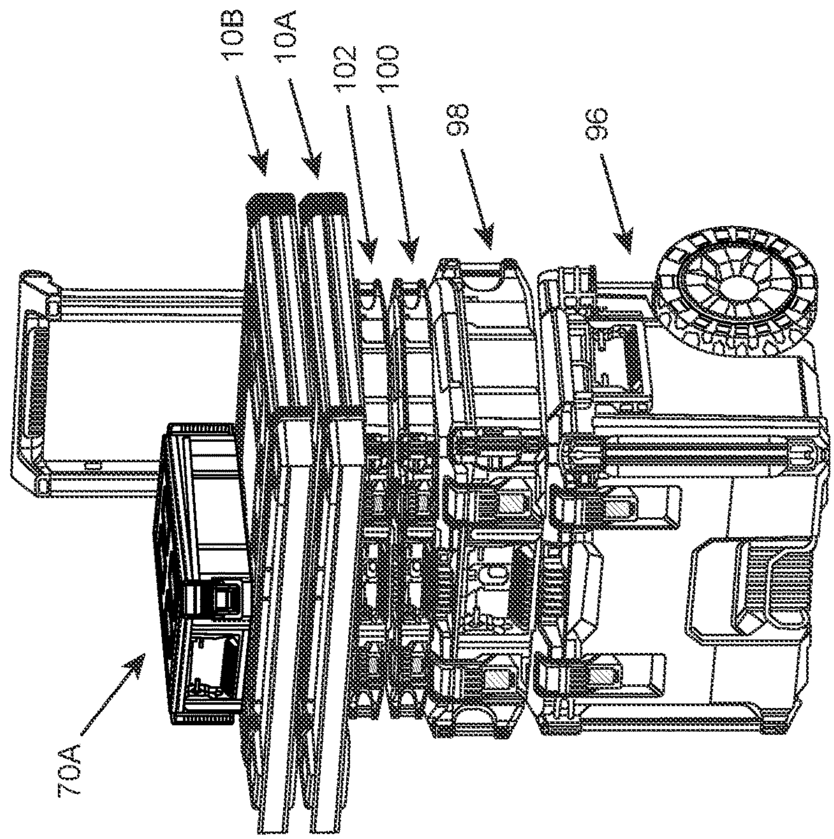

FIG. 11B

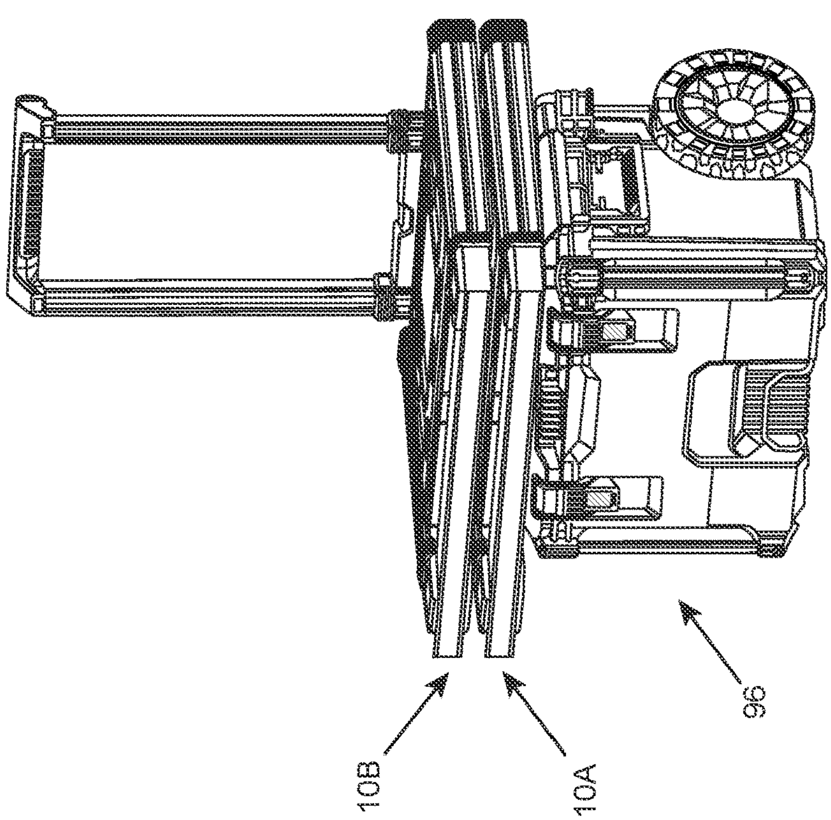

FIG. 11A

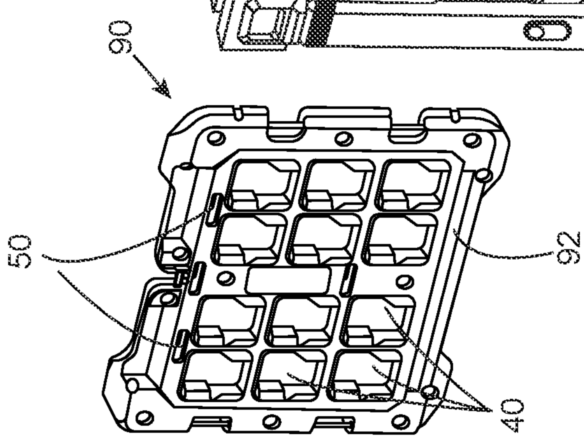

FIG. 12A

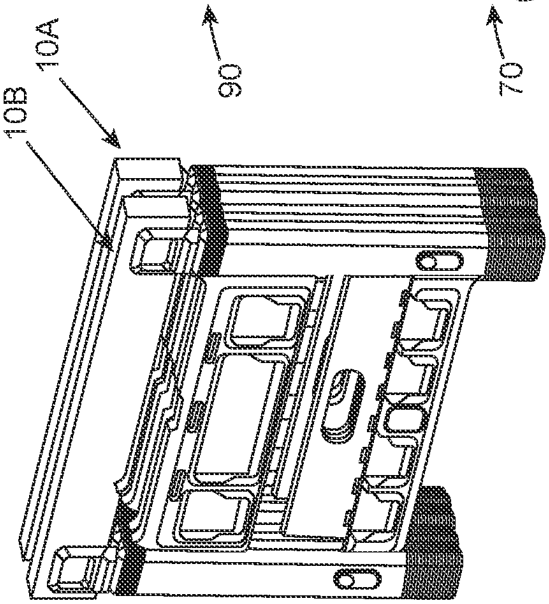

FIG. 12B

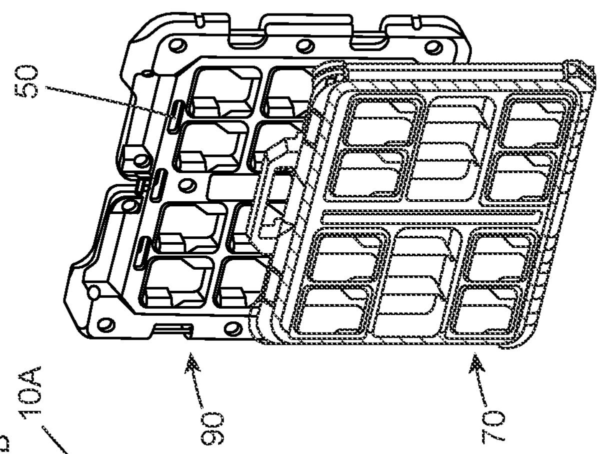

FIG. 12C

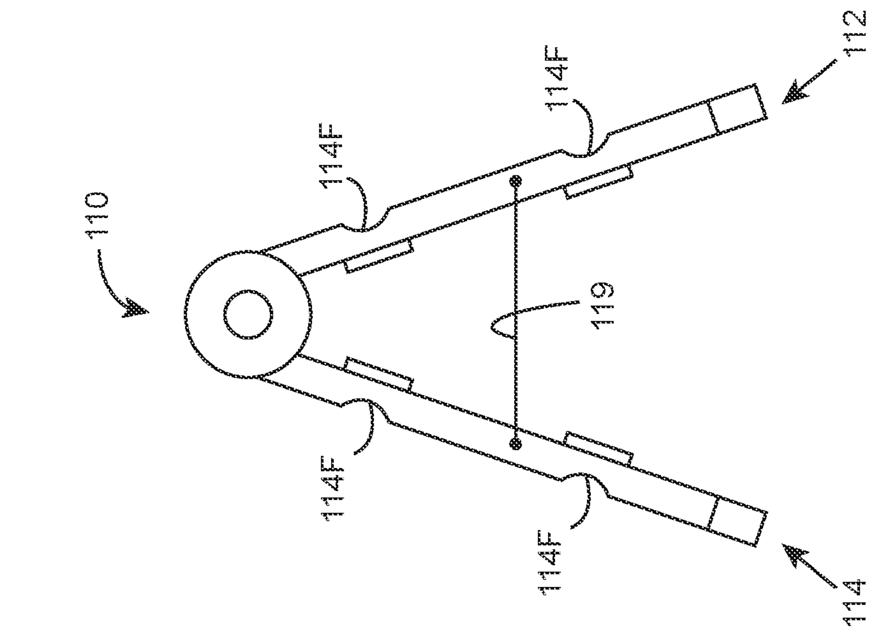

FIG. 13C

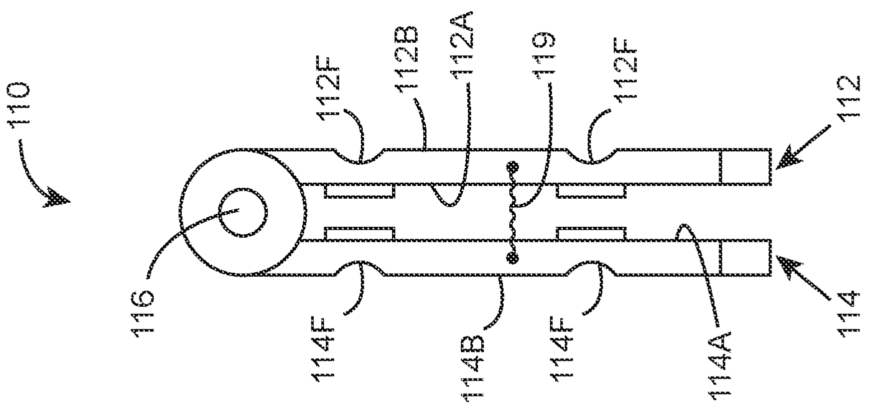

FIG. 13B

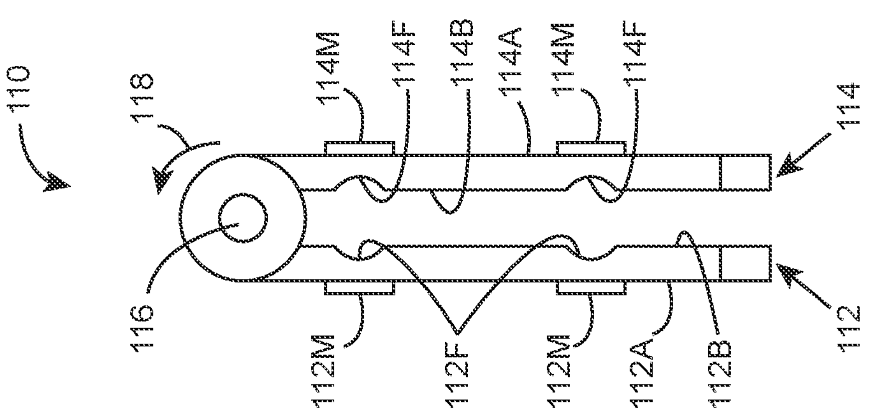

FIG. 13A

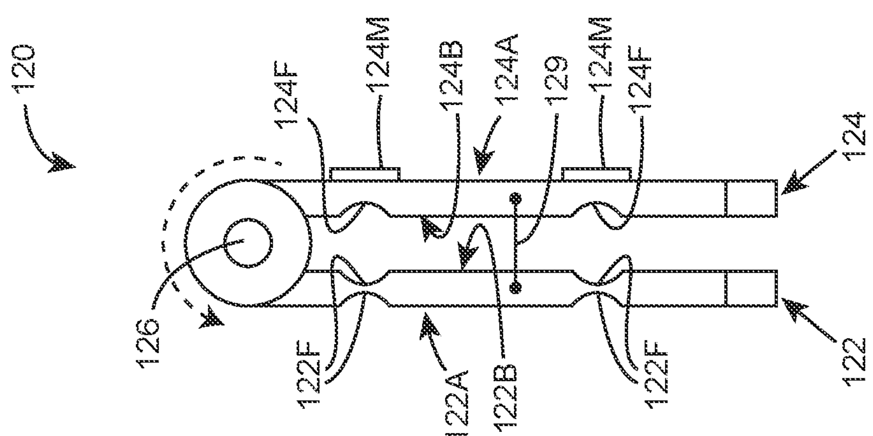

FIG. 14A

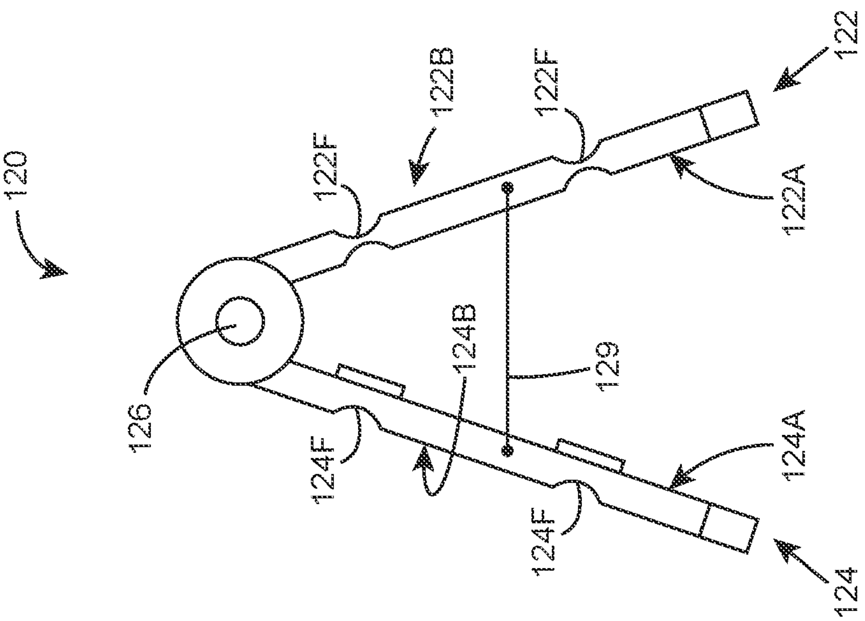

FIG. 14B

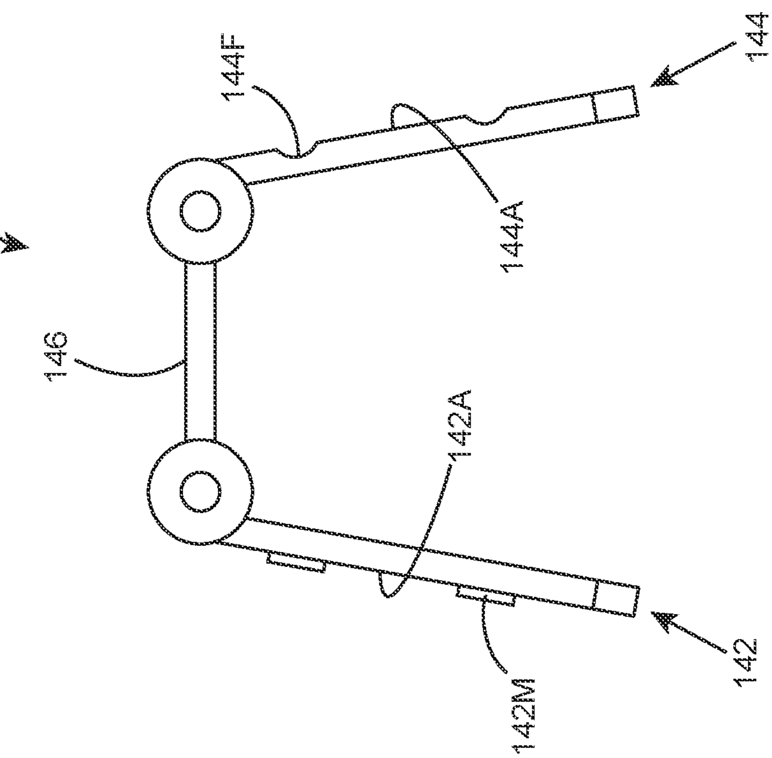

FIG. 15B

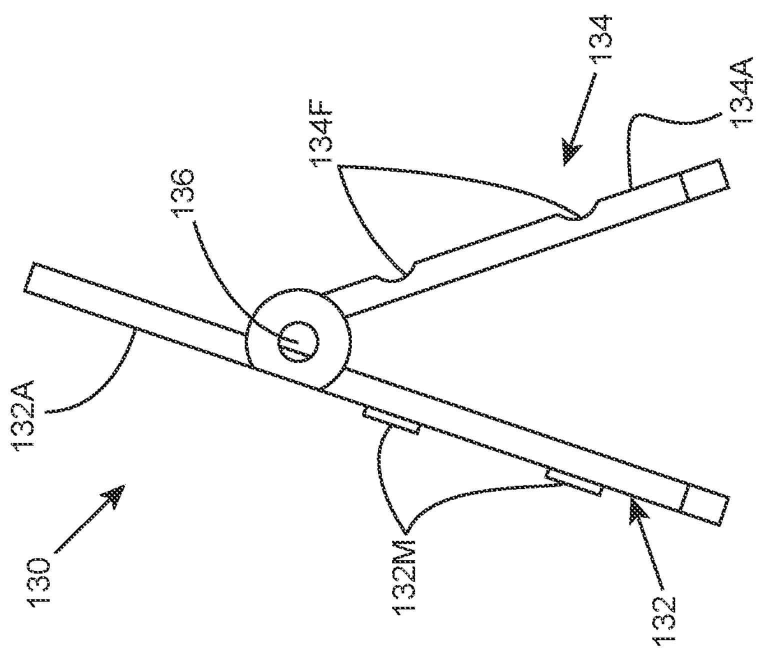

FIG. 15A

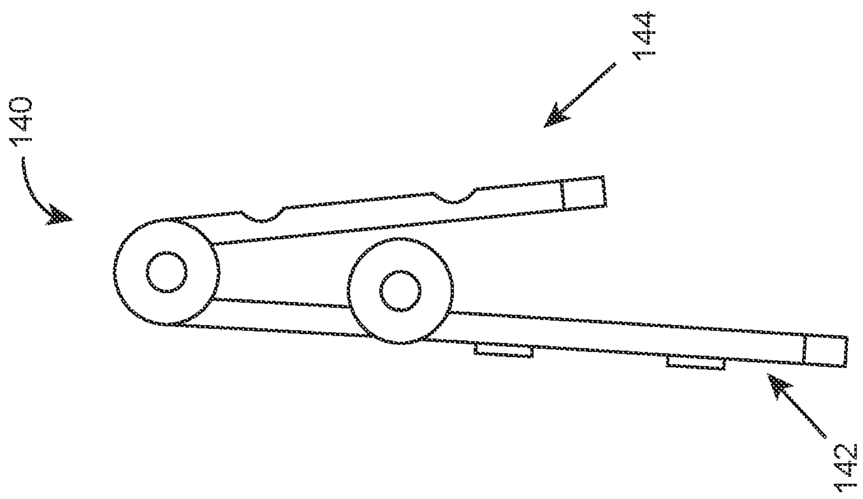

FIG. 15C

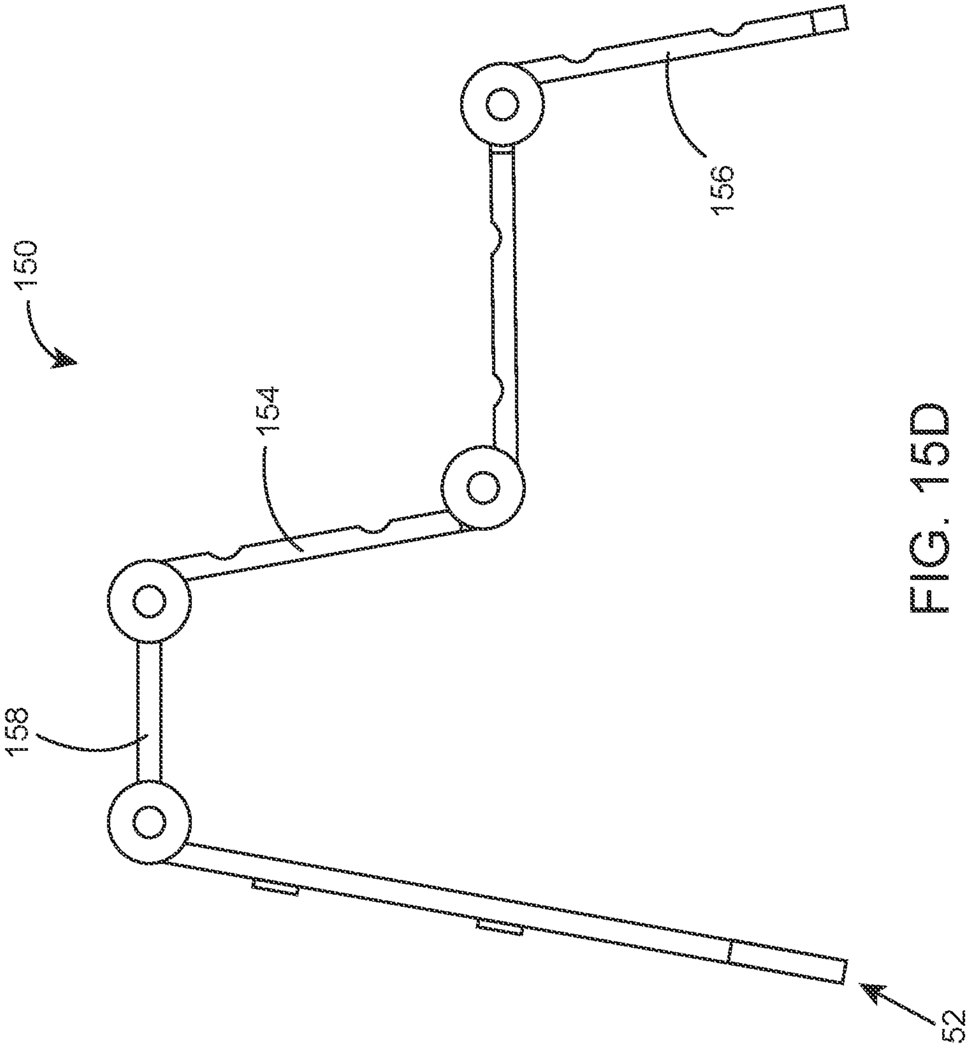

FIG. 15D

SAWHORSE

BENEFIT CLAIM

This application is a United States national stage entry of PCT International Application No. PCT/IL2020/051019 entitled “SAWHORSE”, filed in Israel on Sep. 16, 2020, which claims priority to Israel Patent Application No. 269564, filed on Sep. 23, 2019, the contents of which are herein incorporated in their entirety by reference for all purposes.

TECHNOLOGICAL FIELD

The present disclosure concerns a sawhorse. More particularly the disclosure concerns a sawhorse configured with an articulation system.

Hereinafter in the specification and claims, the term sawhorse is used in its broad sense and denotes a foldable and carriable work station, also referred to in the art as workmate, workbench, worktable, etc.

BACKGROUND ART

References considered to be relevant as background to the presently disclosed subject matter are listed below:

U.S. Pat. No. 9,844,870

U.S. Pat. No. 9,844,870

US 2016/0221177

Acknowledgement of the above references herein is not to be inferred as meaning that these are in any way relevant to the patentability of the presently disclosed subject matter.

BACKGROUND

U.S. Pat. No. 9,844,870 discloses a sawhorse system, comprising: a sawhorse; and a tool holder, having a plurality of sides including a first side, a second side, a third side, and a fourth side, each of the first side and the second side having a free top edge, a bottom edge, a first end edge, and a second end edge, the free top edge of the first side spaced apart from the free top edge of the second side, the third side disposed intermediate and connecting the first end edges of the first side and the second side, and the fourth side disposed intermediate and connecting the second end edges of the first side and the second side, a plurality of pockets attached the plurality of the sides, one of the pockets attached to each of the third side and the fourth side, a first strap directly attached to the free top edge of each of the first side and the second side, and connecting the first side and the second side, a second strap directly attached to the free top edge of each of the first side and the second side, and connecting the first side and the second side, the second strap laterally spaced apart from the first strap, and the free top edges of the first side and the second side are not otherwise connected, a third strap attached at the first end edges of the first side and the second side, the third strap connecting the first side and the second side, wherein the third strap is spaced apart from the one of the pockets on the third side, and an opening is formed between the third strap and the one of the pockets attached to the third side, and the third strap is coplanar with the third side, and a fourth strap attached at the second end edges of the first side and the second side, the fourth strap connecting the first side and the second side, wherein the fourth strap is spaced apart from the one of the pockets on the fourth side, and an opening is formed between the third strap and the one of the pockets attached to the fourth side, and the fourth strap is coplanar with the fourth side, wherein the tool holder is selectively disposed over the sawhorse and is movable between an expanded position and a collapsed position.

U.S. Pat. No. 6,305,498 discloses a sawhorse, comprising a base assembly including a pair of support members each formed from a plastic material, said support members being connected to one another for movement between an open working configuration in which the base assembly is self-supporting in an upright position and a closed storage configuration; a workpiece support structure mounted on said base assembly and defining an upwardly facing support surface for supporting a workpiece when the base assembly is in the self-supporting upright position thereof; and an attachment for said base assembly in the form of a tool carrier assembly comprising a flexible carrier sheet, a plurality of tool holding elements affixed to said carrier sheet and a plurality of fasteners fixedly affixing said carrier sheet to one of said pair of support members in a configuration providing access to tools held by said tool holding elements when said base assembly is in the open working configuration thereof, the tool carrier assembly being positioned (a) such that the tool carrier assembly is below the workpiece support surface to allow the support surface to support a workpiece without interference from said tool carrier assembly and (b) such that said tool carrier assembly does not impede movement of the base assembly between said closed storage configuration and said open working configuration.

US Application 2016/0221177 discloses a sawhorse includes a first side body and a second side body. Each of the first side body and the second side body has a first cross member, a second cross member, a first leg portion with a foot, and a second leg portion with a foot. The first cross member of the first side body is rotatably coupled to the first cross member of the second side body. The first side body and the second side body are positionable between a closed position and an opened position. The sawhorse further includes a centrally hinged, folding platform that is rotatably coupled to the second cross member of each of the first side body and the second side body. Where in the open position, the folding platform is fully deployed to create a substantially planar surface spanning the second cross members of the first side body and the second side body.

GENERAL DESCRIPTION

The present disclosure is directed to a sawhorse comprising a first support frame and a second support frame, each having a first face and a second face, respectively, said second support frame articulated to the first support frame, whereby the sawhorse is manipulable between at least one collapsed position and at least one deployed position; and wherein at least one of the first face and the second face of at least one of the first support frame and the second support frame is configured with an array of articulation locations for detachably attaching one or more utility units to the sawhorse and for detachably attaching the sawhorse to a carrying member.

The term utility unit as used herein denotes any article of utility, either stationary of mobile, having a utility and being detachably attachable to a sawhorse according to the disclosure or to other utility units. A utility unit, according to the present disclosure can be, by way of example, any type of container, work surface, locomotion system, mounting system, hand tool, support element, and the like.

The term carrying member as used herein denotes any surface to which a sawhorse according to the present disclosure can be detachably articulated. By way of example, a carrying member can be a wall surface, a mounting plate, a container wall surface, a cabinet wall surface, and the like.

According to a specific configuration of the disclosure, the articulation location comprises a male-female type coupling arrangement, wherein the sawhorse is configured with one or more male couplers or a female couplers and where the utility unit and carrying member are configured with the other one of male couplers or female couplers.

The sawhorse can further be configured with a locking mechanism associated with the articulation locations, to prevent spontaneous detaching of a utility unit from the sawhorse, i.e. to prevent unintentional separation of the utility unit from a support frame or of the sawhorse from a carrying member.

According to a specific example of the disclosure, there is provided a sawhorse comprising a first support frame and a second support frame pivotally articulated to one another at their respective top portions, each having a first face and a second face, respectively, wherein said first faces face each other and where the sawhorse is manipulable between at least one collapsed position and at least one deployed position; and wherein the second face of at least one of the first support frame and the second support frame is configured with an array of articulating elements (or articulation locations) for detachably attaching one or more utility units to the sawhorse and for detachably attaching the sawhorse to a carrying member.

According to a particular example, the first frame and the second frame are articulated to one another at their respective top portions and wherein at the deployed position the sawhorse has an upside-down V-like shape.

The arrangement can be such that attaching a utility unit over the sawhorse and likewise attaching the sawhorse to a carrying member is easily and readily facilitated by sliding the respective member over the other into engagement with the respective articulation location so that the one or more male-female couplers engage, and the locking mechanism, if applied, snaps into engagement. Detaching a utility unit from the sawhorse and likewise detaching the sawhorse from the carrying member is facilitated by disengaging the locking mechanism, if applied, and sliding the male-female couplers to disengage and away from one another.

Sliding engagement and disengagement take place along a sliding path extending substantially parallel to the male-female couplers members.

Any one or more of the following features, designs and configurations can be applied to the sawhorse according to the present disclosure, separately and in various combinations thereof:

The first support frame and the second support frame can be pivotally articulated to one another at their respective top portions;

The first support frame and the second support frame can be pivotally articulated to one another at their respective top ends;

One of the first support frame and the second support frame can be articulated to the other one of the first support frame and the second support frame at a location between a top end and a bottom end of the other one of the first support frame and the second support frame;

One or both of the first support frame and the second support frame can have a rectangular shape;

The sawhorse can be made of metal or molded resin;

At the collapsed position, the first support frame and the second support frame can extend parallel to one another;

At the collapsed position, facing faces of the first support frame and the second support frame can extend flush against one another;

One or both of the first support frame and the second support frame can comprise at least one height adjusting leg member;

The first support frame and the second support frame can be similar to one another, or identical, and can be interchangeable;

The first support frame and the second support frame can be articulated to one another through one or more interconnecting frame members;

The array of articulation locations can facilitate simultaneous attachment of one or more utility units to a surface of a support frame;

The first support frame and the second support frame can be configured at their top end with integral intervening axle portions for pivotally coupling the first and second support frames to one another;

The array of articulation locations can include one or more such articulation locations, for detachably attaching thereto one or more utility units, and wherein a utility unit can be attached over one or more such articulation locations; i.e. an array of articulation locations can be a set of couplers or a unitary coupling;

Either one or both faces of each of the first support frame and the second support frame can be configured with articulation locations, the articulation locations can be any of male articulation locations and female articulation locations; for example, one face of one of the first support frame and the second support frame can be configured with male articulation locations and another face of the support frame can be configured with male or female articulation locations;

A face of a support frame can be configured with a combination of male and/or female articulation locations;

The first support frame and the second support frame can be swingable between a collapsed position and at least one deployed position;

At least one of the first support frame and the second support frame can be swingable about the other one of the first support frame and the second support frame substantially about 360°;

A top end of the sawhorse can be configured with a utility portion; the utility portion can be a workpiece grip, a vise, a top bar, etc.;

The locking mechanism can be a locking latch extending from one of a support frame and a utility unit or carrying member, and configured for arresting within a locking receptacle within the other one of a support frame and a utility unit or carrying member;

The locking latch can be configured for spontaneous snap engagement within the locking receptacle;

The articulation locations can be integral with the sawhorse frame, or integrated therewith;

One or more articulation locations can be disposed over a side portion of the first support frame and the second support frame;

The sawhorse can be configured with a span-restricting member, for restricting angular displacement between the first support frame and the second support frame;

A span-restricting member can be a support member configured with one or more articulating locations;

According to one particular arrangement, a face of at least one of the first support frame and the second support frame defines an articulation face, and the articulation locations comprise at least one male coupler at one of the articulation face and a face of the utility unit, and a female coupler at the other one of the articulation face and the face of the utility unit, said female coupler having a depressed locking location configured with at least one locking rib extending above a depressed plane and along a sliding path, and having an open edge facing in a first sense; said male coupler having a projecting locking location disposed in register with said depressed locking location and configured with at least one locking tongue extending along said engaging sliding path at a second sense, opposite to said first sense, and configured for arresting engagement at a space between said locking rib and depressed plane;

The at least one locking rib of the articulation locations can extend substantially parallel to the sliding path;

The at least one locking rib can extend substantially perpendicular and intersect the sliding path;

The female coupler can comprise a single locking rib extending at rear end of the depressed locking location and substantially perpendicular to the sliding path;

The female coupler can comprise two locking ribs extending at side edges of the depressed locking location and disposed substantially parallel to the sliding path;

The female coupler can comprise two locking ribs each extending at a respective side edge of two neighboring depressed locking locations, said locking ribs disposed substantially parallel to the sliding path;

The coupling arrangement can be configured such that engaging the utility unit with the articulation face of the sawhorse is facilitated by sliding the utility unit with respect to the utility unit articulation face along a sliding path defined by at least one of said at least one locking rib and said at least one locking tongue;

The coupling arrangement can be configured for snap-type locking of a utility unit over the a utility unit articulation face;

The locking mechanism can comprise at one of the articulation face and a face of the utility unit, a locking latch arresting location, and the other one of said articulation face and a face of the utility unit a locking member displaceable and disposed in register with said locking latch arresting location, wherein at a locked position the locking member is arrested by the corresponding locking latch arresting location, and further wherein disengaging the utility unit from the sawhorse is facilitated by disengaging the at least one locking latch from the at least one locking latch arresting location;

The locking mechanism can further comprise a release latch for displacing the locking latch into disengagement from the locking latch arresting location;

The locking mechanism can be spring biased and configured for normally projecting from a face of the utility unit or from the articulation face of the sawhorse;

The locking latch can be associated with and extend from a male coupler element.

BRIEF DESCRIPTION OF THE DRAWINGS

In order to better understand the subject matter that is disclosed herein and to exemplify how it may be carried out in practice, embodiments will now be described, by way of non-limiting examples only, with reference to the accompanying drawings, in which:

FIG. **1**A is a perspective view of a sawhorse according to an example of the present disclosure with female articulation locations, the sawhorse at its collapsed position;

FIG. **1**B is a perspective view of the sawhorse of FIG. **1**A with male articulation locations;

FIG. **1**C is a side view in direction of arrow I in FIG. **1**B;

FIG. **1**D is a section taken through a top portion of the sawhorse, along line I-I in FIG. **1**B;

FIG. **2**A is a planer view of the female side of the sawhorse;

FIG. **2**B is a planer view of the male side of the sawhorse;

FIG. **3**A is a side perspective view of the sawhorse at a deployed, extended position with female articulation locations;

FIG. **3**B is a male side perspective view of the sawhorse of FIG. **3**A;

FIG. **4**A is an enlargement of the portion marked **4**A in FIG. **3**A;

FIG. **4**B is an enlargement of the portion marked **4**B in FIG. **3**B;

FIGS. **5**A and **5**B are locally sectioned views along line II-II in FIG. **10**A, illustrating consecutive steps of interlocking a utility unit container to the sawhorse, by an articulation arrangement according to the disclosure;

FIGS. **6**A and **6**B are locally sectioned views along line III-III in FIG. **10**A, illustrating consecutive steps of locking a utility unit container to the sawhorse, according to the disclosure;

FIG. **7** is a bottom perspective view of a container configured with male couplers according to an aspect of the disclosure;

FIG. **8**A is an enlarged front perspective view of a male coupler;

FIG. **8**B is an enlarged rear perspective view of the male coupler;

FIG. **9**A is a perspective view of two sawhorses according to the disclosure articulated to one another and to a carrier platform, according to an example of the disclosure;

FIG. **9**B is a horizontal section along line IX-IX in FIG. **9**A;

FIG. **9**C is an enlargement of the portion marked **9**C in FIG. **9**B;

FIGS. **10**A and **10**B illustrate examples of articulating utility units to the sawhorse;

FIGS. **11**A and **11**B are examples of articulating the sawhorse to a wheeled container assembly;

FIG. **12**A is a perspective view of a mounting platform used in conjunction with a sawhorse according to the disclosure;

FIG. **12**B illustrates two sawhorses mounted on a mounting platform according to FIG. **12**A;

FIG. **12**C illustrates a container mounted on a mounting platform according to FIG. **12**A;

FIGS. **13**A to **13**C are schematic side views of a sawhorse according to a configuration of the disclosure, at a collapsed position, an inverted collapsed position and a deployed position, respectively;

FIGS. **14**A and **14**B are yet an example of a sawhorse according to a configuration of the disclosure, at a collapsed position and an inverted collapsed position; and

FIGS. **15**A to **15**D are different examples of sawhorses according to other configurations of the disclosure.

DETAILED DESCRIPTION OF EMBODIMENTS

Attention is directed first to FIGS. **1**A to **3**B, illustrating a sawhorse according to the present disclosure, generally designated **10**. The sawhorse **10** comprises a first support frame **12** and a second support frame **14**, both having a similar, generally rectangle shape, and each having a first face **12**A and **14**A and a second face **12**B and **14**B, respectively. For sake of example and clarification, in the present example the first face **12**A and **14**A are referred to as outside faces of the frames (i.e. facing away from each other) and the second face **12**B and **14**B are referred to as inside faces of the frames (i.e. facing towards each other). The first support frame **12** and the second support frame **14** are pivotally articulated to one another at their respective top ends at a pivot axle **18**, along an axis X (best seen in FIG. **1**D), such that the sawhorse **10** is configurable between a collapsed/folded position (FIGS. **1**A to **1**D) and a deployed/expanded position (FIGS. **3**A and **3**B).

The sawhorse **10** is further configured with a span-restricting arrangement comprising a first pivotal shelf-like member **15** extending from a first, inside face **12**B of the first support frame **12**, and a second, oppositely extending pivotal shelf-like member **17** extending from a second, inside face **14**B of the second support frame **14**. Shelf-like members **15** and **17** are pivotally articulated to one another at **19**, and configured for restricting angular displacement between the first support frame **12** and the second support frame **14**, so as to retain the sawhorse at a fixed and stable deployed position. At the deployed position, shelf-like members **15** and **17** extend coplanar and serve together as a shelf member. The restricting shelf-like members **15** and **17** displace into the coplanar shelf-like position upon deploying the sawhorse into its operable, open/deployed position. Each of the shelf-like member **15** and **17** is further configured with an opening **21**, which at the collapsed position are in register with one another, serving together as a carrying handle. However, it is appreciated that other restricting arrangements can be configured, such as a strap or a sling extending between the first support frame **12** and the second support frame **14**.

Extending at a top of each side end of the sawhorse **10**, there are one or more U-like workpiece supports **20**, in this example two workpiece supports **20** coextending and supporting a rectangular support beam **22** which can be integral with the two workpiece supports **20** or articulated thereto through fasteners **24** (better seen in FIG. **1**D), rendering it replaceable.

The first support frame **12** and the second support frame **14** comprise each two side leg members **26** and **28**. Each side leg member **26** and **28** is configured with an extension leg portion **30** and **32**, respectively, telescopically received within the respective side leg members **26** and **28**, and each comprising at a bottom end thereof an anti-slip shoe **36**. The extension leg portions **30** and **32** are selectively arrested at a desired level within the respective side leg members **26** and **28** by any one of several snap tabs **38** configured for protecting and arresting through an arresting opening **39** formed at the side leg members **26** and **28**.

In the present example, the first face **12**A of the first support frame **12** (namely the outside face thereof) is configured with an array of female-type articulation locations **40**A, **40**B and **40**C (being of different size though having the same arresting arrangement as will be discussed), as will be discussed hereinafter in detail.

Further in present example, the first face **14**A of the second support frame **14** (namely the outside face thereof) is configured with an array of male-type articulation locations **60**, as will be discussed hereinafter in detail.

With particular reference to FIG. **4**A, the female-type articulation locations **40**A, **40**B and **40**C (collectively designated **40**) each comprise a depressed locking surface **42** configured with a pair of parallelly disposed locking ribs **44** facing towards one another and extending above the depressed surface **42** and along a sliding path **48** (which is the orientation at which a utility unit is displaced over during articulation, wherein **48**A is the locking direction and **48**B is the unlocking direction), and having a slanted open edge **52** facing in a first sense along the sliding path **48**. An outside surface of the locking ribs **44** is substantially flush with a sliding surface of the face **12**A; a space **53** (FIGS. **5**A and **5**B) extends between the locking ribs **44** and the depressed surface **42** and it is seen that a bottom face of the locking ribs **44** extends substantially parallel to said depressed surface **42**.

The outside, first face **12**A of the first support frame **12** is further configured with at least one locking latch arresting location, i.e. recess **50**, the purpose of which will be discussed below.

Reverting now to the male-type articulation locations **60** configured at the outside face **14**A of the second support frame **14**, and disposed also at a bottom face of a utility unit, for example the container **70** of FIG. **7**. The male-type articulation locations, namely couplers **60**, seen at larger scale in FIGS. **8**A and **8**B, extend in register with the female couplers **40** of the sawhorse **10** (and of a mounting plate **90** seen in FIG. **12**A), as far as shape, size and positioning, whereby a utility unit can be articulated to the sawhorse at several locations thereof. Each male coupler **60** comprises a projecting locking location **68** disposed in register with said depressed female-type locking location **40**, and is configured with a pair of laterally extending locking tongues **72** one at each side of the projecting locking location **68**, i.e. facing away from one another, laterally extending along said engaging sliding path **48** (at a second, sense, opposite to said first sense), and configured for arresting engagement at the space **53** between said locking rib **44** and depressed surface **42**.

Further noted, the utility unit articulation face of the sawhorse, namely face **12**A thereof (as well as mounting plate **90** of FIG. **12**A) is configured with at least one locking latch arresting location, i.e. recesses **50**, and the male face of the sawhorse, namely face **14**A, as well as a bottom face **71** of utility unit (container **70**) is configured with a locking latch **82** (FIGS. **6**A and **6**B), spring biased by spring **86** to project from the face **14**A of the sawhorse **10** and from the bottom face **71** of the container **70**, and being manually displaceable by a manipulating grip **87** through an aperture or recess **89** (seen also in FIG. **4**B), between the normally projecting position and a manually retracted position (FIG. **6**A), said locking member **82** disposed in register with at least one of the locking latch arresting locations **80**.

When it is required to detachably attach a utility unit to the sawhorse **10**, the former is placed over the later, placing the projecting locking portions **68** within the depressed locking location **42** with the locking tongues **72** slidingly engaging below and being gradually arrested by the respective locking ribs **44** until complete arresting is obtained (FIG. **5**B). At the fully engaged locked position (FIG. **5**B) each locking rib **44** is disposed between the respective locking tongue **72** and the bottom face **62** of the utility unit (container **70**), however bearing flush against at least a portion of the top surface of the locking tongues **72**. Simultaneously, as the locking tongues **72** arrest by the locking ribs **44**, the locking latch **82** of the utility unit (container **70**) slides into the locking latch arresting location (recess **50**) and finally snaps into locking engagement therewith.

It is noted the locking latch **82** is configured with a chamfered nose **91** whereby upon sliding displacement of the utility unit along the sliding path **48**A it first glides over a guiding ramp surface **93** of the sawhorse, displacing it upwards against the biasing effect of spring **89** (in direction of arrow **95** in FIG. **6**A), whereby when the utility unit **70** reaches the final locking position over the respective articulation face of the sawhorse, the locking latch **82** plunges into arresting position within the recess **50**.

At the locked position, utility unit **70** is attached over the articulation face of the respective support frame **12**A of the sawhorse. Detaching of a utility unit **70** from a sawhorse **10** is easily facilitated by unlocking, obtained by displacing the locking latch **82** upwards against the biasing effect of spring **86**, whereby the utility unit **60** can be slidingly displaced along path **48**B, and detached.

It is appreciated that the container **70** is a mere example of a utility unit, which can be any article of utility, either stationary of mobile, having a utility and being detachable attachable to other utility units. A utility unit according to the present disclosure can be, by way of example, any type of container, work surface, locomotion system, mounting system, hand tool, support element, and the like.

The male/female articulation arrangement disclosed herein is an example, and is understood that other coupling arrangements can be utilized, such as, for example, a bayonet coupling arrangement, however bearing in mind that the coupling arrangement should offer a fast and reliable attaching/detaching arrangement between the utility unit and the sawhorse, and for attaching the sawhorse to a carrying member. #

Turning now to FIGS. **9**A to **9**C and **12**B there is illustrated an arrangement wherein a first sawhorse according to the present disclosure, generally designated **10**A, is detachably articulated to a mounting plate **90** similar to that disclosed in FIG. **12**A, and wherein a second sawhorse according to the present disclosure, generally designated **10**B, is detachably articulated to first sawhorse **10**A. The arrangement is such that the face **92** of the mounting plate **90** is configured with a plurality of female-type articulation locations **40** (similar to female-type coupling locations **40** configured on the first face **12**A of the sawhorse **10** as disclosed hereinbefore), whereby the first sawhorse **10**A is slidingly articulated over the mounting plate **90** by male type couplers disposed over the first face **14**A of the first sawhorse **10**A slidingly articulated with said female articulation locations **40** of the mounting plate **90**. In turn the second sawhorse **10**B is detachably mounted over the first sawhorse **10**A through the male-type articulation locations **60** at the first face **14**A of the second sawhorse **10**B being slidingly articulated with the female-type articulation locations **40** at the first face **12**A of the first saw horse **10**A.

It is appreciated that in FIGS. **12**A to **12**C that the mounting plate **90** serves as a carrying member to which a sawhorse **10** can be detachably articulated, wherein the mounting plate **90** can be integral with or integrated with a carrying wall surface, etc. Also, the mounting plate **90** can be used for directly attaching thereto various utility units (FIG. **12**C), such as container **70** (as disclosed in connection with FIG. **7**).

FIG. **10**A illustrates a sawhorse **10** according to the present disclosure, and as disclosed hereinabove, at its deployed, open position, wherein an organizer container **70**A is detachably articulated over the first face **12**A of the first support frame of the sawhorse **10**. Also attached to the first face **12**A there is a holder **97** useful, for example, for supporting a hand tool such as a drill, hammer, jigsaw, etc.

It is appreciated that both the organizer container **70**A and the holder **97** are each configured with one or more male type-couplers adapted for engagement with corresponding female-type articulation locations **40** at the first face **12**A of the sawhorse **10**.

In FIG. **11**A there is illustrated a pair of sawhorses **10**A and **10**B, according to the disclosure hereinbefore, detachably secured to one another (similar to the arrangement disclosed in connection with FIG. **9**) and in turn being detachably articulated over a top cover of a wheeled container **96**, wherein articulation is facilitated through the first face **14**A of the second support frame **14** of the sawhorse **10**, i.e. the male-type articulation locations **60** thereof being slidingly engaged with female-type articulation locations disposed over the cover of the container **96**, said female-type articulation locations being similar to articulation locations **40** of the sawhorse **10**.

In FIG. **11**B, the pair of sawhorses **10**A and **10**B is mounted over an organizer container **102** which in turn is articulated over another organizer container **100**, and which in turn is articulated over a container **98** and in turn over the wheeled container **96**, however with a smaller container **70**A articulated over the female-type couplers **40** at the first face **12**A of the top sawhorse **10**B. it is appreciated that coupling between all containers and sawhorses is facilitated through a male-female type articulation locations as disclosed hereinabove, with a locking arrangement provided (as in connection with FIGS. **6**A and **6**B).

In FIGS. **13**A to **14**B there are schematic illustration of two variations of a sawhorse according to the present disclosure, both following the same principals. The sawhorse **110** of FIGS. **13**A to **13**C comprises a first support frame **112** and a second support frame **114** pivotally articulated to one another at their top end, about axis **116**. The first support frame **112** has a first face **112**A configured with an array of male-type coupling locations **112**M and a second face **112**B configured with an array of female-type coupling locations **112**F. Likewise, the second support frame **114** has a first face **114**A configured with an array of male-type coupling locations **114**M and a second face **114**B configured with an array of female-type coupling locations **114**F. In the position of FIG. **13**A all male-type couplers **112**M and **114**M are disposed on the outside faces of the sawhorse **110**. However, upon pivotally swinging the second support frame **114** about the axis **116** through 360° in direction of arrow **118** in FIG. **13**A, the new position is such that all outside faces of the sawhorse are now positioned with female-type couplers **112**F and **114**F, respectively such that at a deployed, working position (FIG. **13**C) there is more surface area configured with female-type couplers, for articulating more utility units (not shown). It is noted that angular restricting member **119** is configured for securing the angular opening between the support frames.

The sawhorse **120** of FIGS. **14**A and **14**B comprises a first support frame **122** and a second support frame **124** pivotally articulated to one another at their top end, about axis **126**. The first support frame **122** has a first face **122**A and a second face **122**B, both configured with an array of female-type coupling locations **122**F. The second support frame **124** has a first face **124**A configured with an array of male-type coupling locations **124**M and a second face **124**B configured with an array of female-type coupling locations **124**F. In the position of FIG. **14**A the sawhorse can be disposed for articulating the face **124**A over a mounting plate (not shown) whilst the face **122**A can serve for articulating thereto different utility units (not shown). Notably, the sawhorse **120** can be deployed at this situation as well. However, in the position of FIG. **14**B, the sawhorse second support frame **124** is swung by 360° over axis **126**, whereby only female-type couplers **122**F and **124**F are disposed at the outside faces **122**B and **124**A, respectively, providing for increased mounting area over the sawhorse **120**.

FIGS. **15**A to **15**D schematically illustrate examples of different sawhorse configurations according to the present disclosure. In FIG. **15**A the sawhorse **130** has a first support frame **132** and a second support frame **134** pivotally articulated to the first support frame **132** at a pivot axis **136** disposed intermediately along the first support frame. The first support frame **132** is configured at a first face thereof **132**A with a plurality of male-type couplers **132**M, and the second support frame **134** is configured at a first face thereof **134**A with a plurality of female-type couplers **134**F.

The sawhorse **140** of FIGS. **15**B and **15**C comprises a first support frame **142** and a second support frame **144**, pivotally articulated at their respective top end to an interconnecting support link **146**, wherein the first support frame **142** is configured at a first face thereof **142**A with a plurality of male-type couplers **142**M, and the second support frame **144** is configured at a first face thereof **144**A with a plurality of female-type couplers **144**F. the sawhorse is configurable between a deployed, table-like configuration (FIG. **15**B) and a collapsed, stowing position (FIG. **15**C).

The sawhorse **150** of FIG. **15**D resembles that disclosed in FIGS. **15**B and **15**C, comprises a first support frame **152** and an interconnection top link frame **158**, however the second support frame is composed of two pivotally articulated segments **154** and **156**, imparting the sawhorse additional modularity, and wherein at least one face of any of the support frames is configured with coupling locations (male and/or female couplings as discussed above).

The invention claimed is:

1. A sawhorse comprising

a first support frame and a second support frame, each having

a first surface and a second surface, respectively, said second support frame articulated to the first support frame,

whereby the sawhorse is manipulable between at least one collapsed position and at least one deployed position; and

wherein at least one of the first surface and the second surface of at least one of the first support frame and the second support frame is configured with an array of articulation locations for detachably attaching one or more utility units to the sawhorse and for detachably attaching the sawhorse to a carrying member, wherein

a face of at least one of the first support frame and the second support frame defines an articulation face, and the articulation locations comprise at least one female coupler, said female coupler having

a depressed locking location configured with at least one locking rib extending above a depressed plane and along a sliding path, and

having a slanted open edge facing in a first direction along the sliding path; the female coupler being configured for detachably attaching thereto a male coupler defined on a face of the utility unit, said male coupler having a projecting locking location disposable in register with said depressed locking location and configured with at least one locking tongue extending along an engaging sliding path at a second direction, opposite to said first direction, and configured for arresting engagement with the female coupler at a space between said locking rib and depressed plane of said female coupler.

2. The sawhorse of claim **1**, wherein the sawhorse is configured with a plurality of said female couplers.

3. The sawhorse of claim **1**, wherein sliding engagement and disengagement of a utility unit take place along a sliding path extending substantially parallel to the depressed plane of the female couplers.

4. The sawhorse claim **1**, further comprising a locking mechanism associated with the articulation locations configured to prevent spontaneous detaching of a utility unit from the sawhorse.

5. The sawhorse of claim **4**, wherein the locking mechanism comprises a locking latch arresting location located at the articulation face and configured for engagement with a locking member displaceable and disposable in register with said locking latch arresting location of the utility unit, wherein at a locked position the locking latch arresting location is configured to arrest the locking member, and further wherein disengaging the utility unit from the sawhorse is facilitated by disengaging the locking latch from the locking latch arresting location.

6. The sawhorse of claim **5**, wherein the locking mechanism is spring biased.

7. The sawhorse of claim **5**, wherein the locking latch is associated with and extends from a male coupler.

8. The sawhorse of claim **1**, wherein the first frame and the second frame are articulated to one another at their respective top portions and wherein at the deployed position the sawhorse has an upside-down V-like shape.

9. The sawhorse of claim **1**, wherein one or both of the first support frame and the second support frame comprise a height adjusting leg member.

10. The sawhorse of claim **1**, wherein the first support frame and the second support frame are articulated to one another through one or more interconnecting frame members.

11. The sawhorse of claim **1**, wherein both faces of each of the first support frame and the second support frame is configured with said articulation locations.

12. The sawhorse of claim **1**, wherein the first support frame and the second support frame are swingable between a collapsed position and at least one deployed position.

13. The sawhorse of claim **1**, wherein the articulation locations are integral with the first support frame and the second support frame, or integrated therewith.

14. The sawhorse of claim **1**, wherein the at least one locking rib of the articulation locations extends substantially parallel to the sliding path.

15. The sawhorse of claim **1**, wherein the at least one locking rib extends substantially perpendicular and intersect an interlocking sliding path.

16. The sawhorse of claim **1**, wherein the female coupler comprises a single locking rib extending at rear end of the depressed locking location and substantially perpendicular to the sliding path.

17. The sawhorse of claim **1**, wherein the female coupler comprises two locking ribs extending at side edges of the depressed locking location and disposed substantially parallel to the sliding path.

18. The sawhorse of claim **1**, wherein the female coupler comprises two locking ribs each extending at a respective side edge of two neighboring depressed locking locations, said locking ribs disposed substantially parallel to the sliding path.

19. The sawhorse of claim **1**, wherein the articulation location is configured such that engaging the utility unit with the articulation face of the sawhorse is facilitated by sliding the utility unit with respect to the utility unit articulation face along a sliding path defined by at least one of said at least one locking rib and said at least one locking tongue.

20. The sawhorse of claim **4**, wherein the locking mechanism comprises a locking member, configured for displacing into a locking latch arresting location located at a face of the utility unit, wherein at a locked position the locking member is configured to be arrested within the locking latch arresting location.

21. A sawhorse assembly comprising a sawhorse and one or more utility units,

the sawhorse comprising

a first support frame and a second support frame, each having a first surface and a second surface, respectively, said second support frame articulated to the first support frame, whereby the sawhorse is manipulable between at least one collapsed position and at least one deployed position; and wherein at least one of the first surface and the second surface of at least one of the first support frame and the second support frame is configured with an array of articulation locations for detachably attaching said one or more utility units to the sawhorse and for detachably attaching the sawhorse to a carrying member, and

wherein a face of at least one of the first support frame and the second support frame defines an articulation face, and

the articulation locations comprise at least one male coupler at one of the articulation face and a face of the utility unit, and a female coupler at the other one of the articulation face and the face of the utility unit, said female coupler having

a depressed locking location configured with at least one locking rib extending above a depressed plane and along a sliding path, and

having a slanted open edge facing in a first direction along the sliding path; said male coupler having a projecting locking location disposed in register with said depressed locking location and configured with at least one locking tongue extending along an engaging sliding path at a second direction, opposite to said first direction, and configured for arresting engagement with the female coupler at a space between said locking rib and depressed plane of said female coupler.

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